The Era of Large Models and Expanding Horizons

– Applications and Challenges –

Elliott Wu

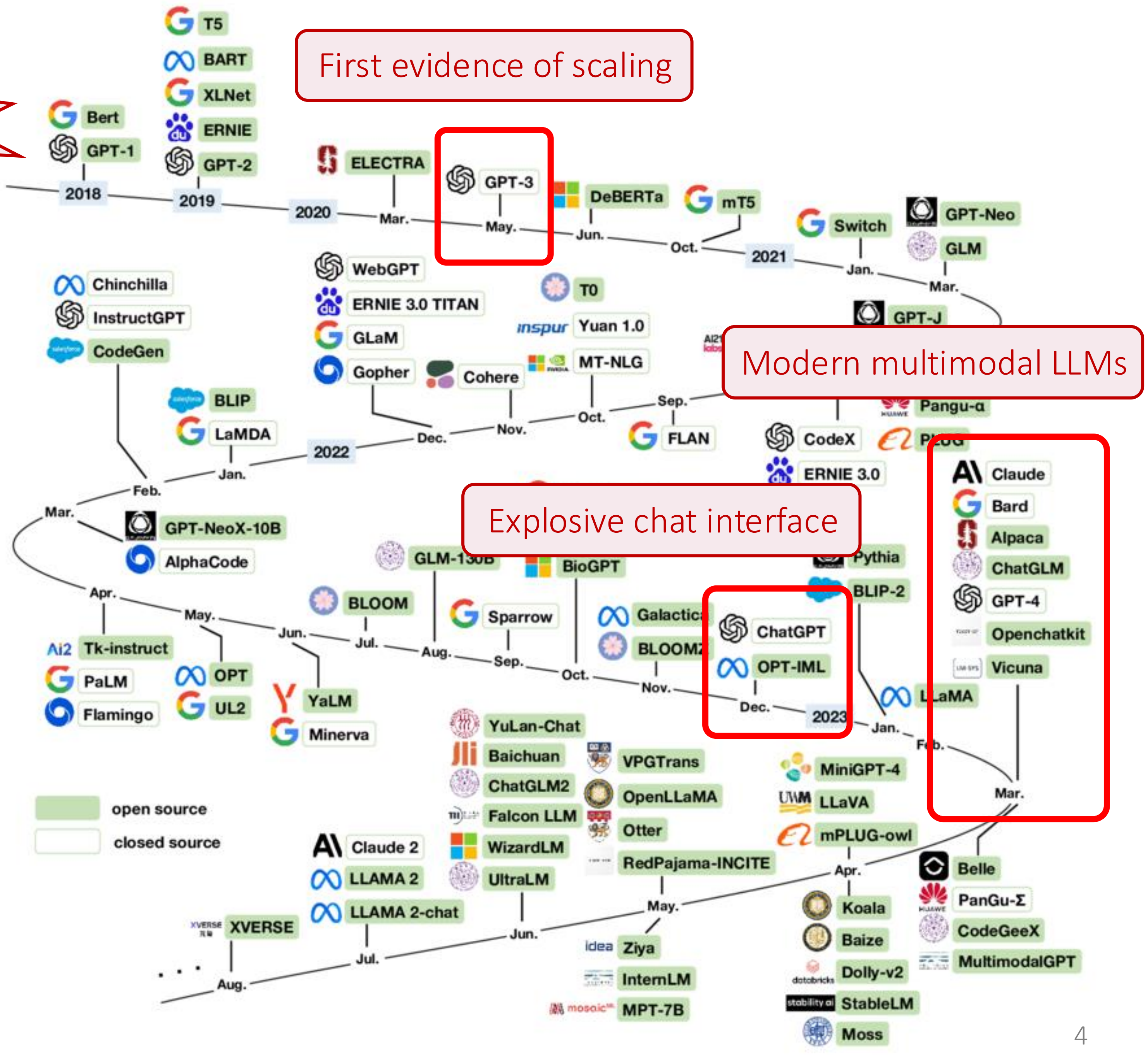
Applications and Challenges – Outline

- Part 1 – The Era of Large Models
- Part 2 – Expanding Horizons

Part 1 – The Era of Large Models

Transformers
(2017)

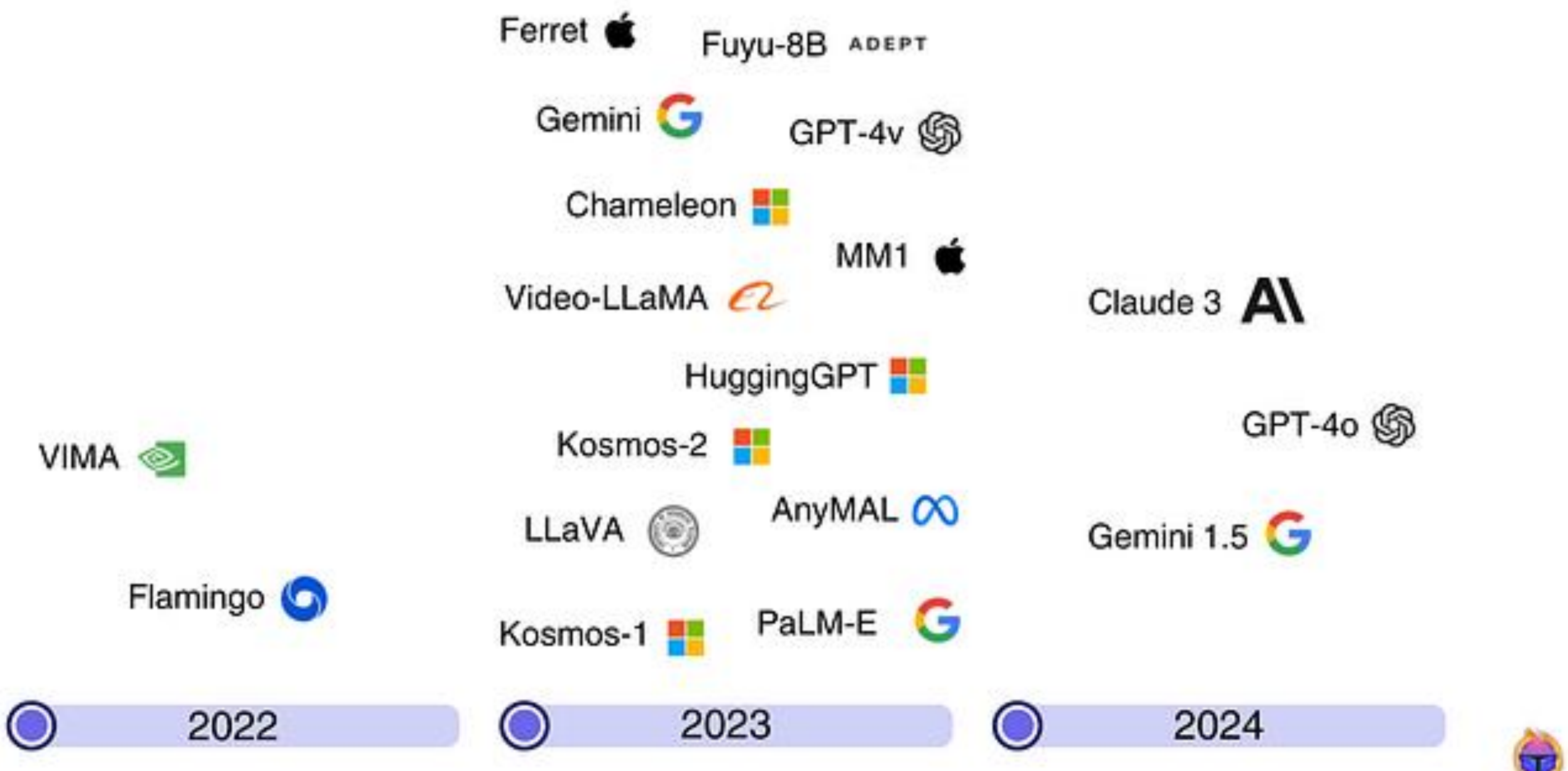
A Brief History of LLMs (2018-2023)



- GPT-1 (2018): 117M params
- GPT-2 (2019): 1.5B params
- GPT-3 (2020): 175B params
- GPT-4 (2023): >1T params

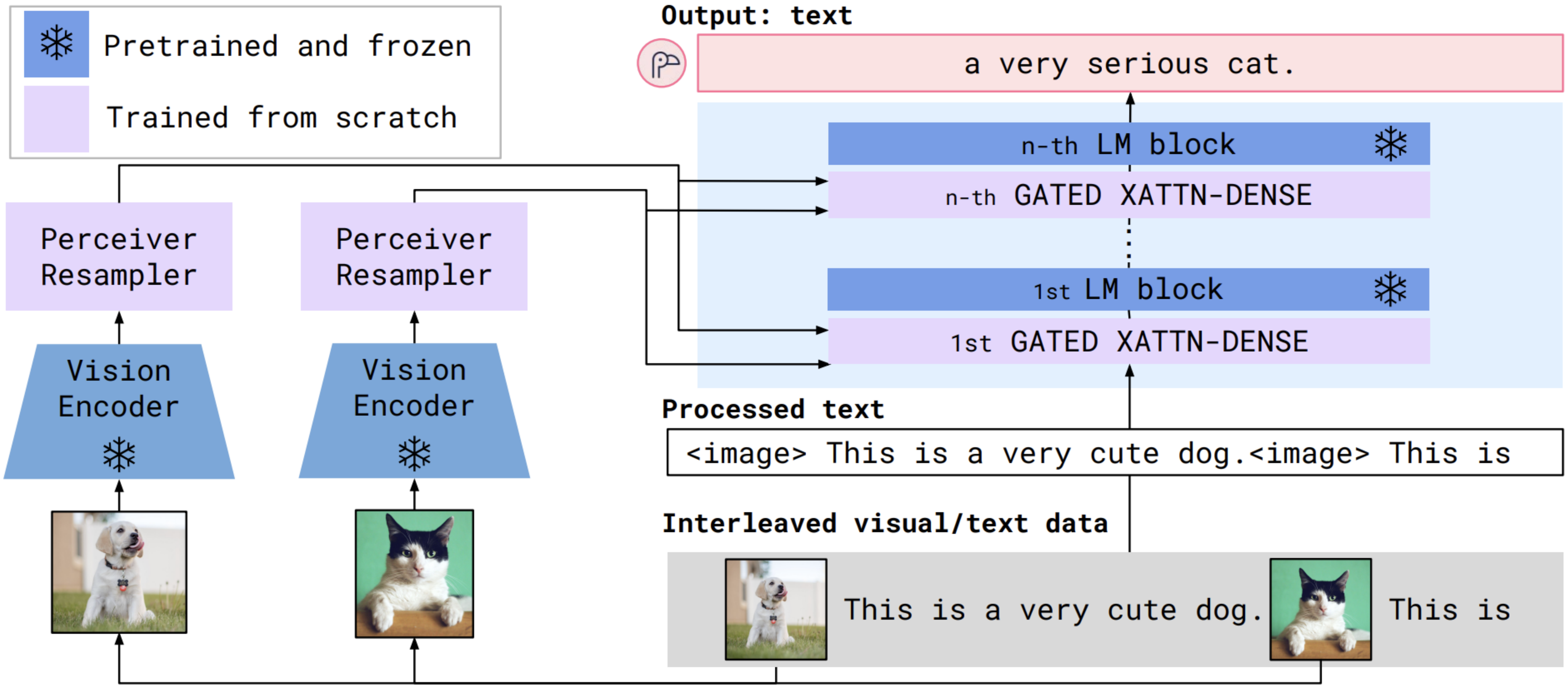
https://github.com/GPT-Alternatives/gpt_alternatives

Multimodal Language Models (2022-2024)

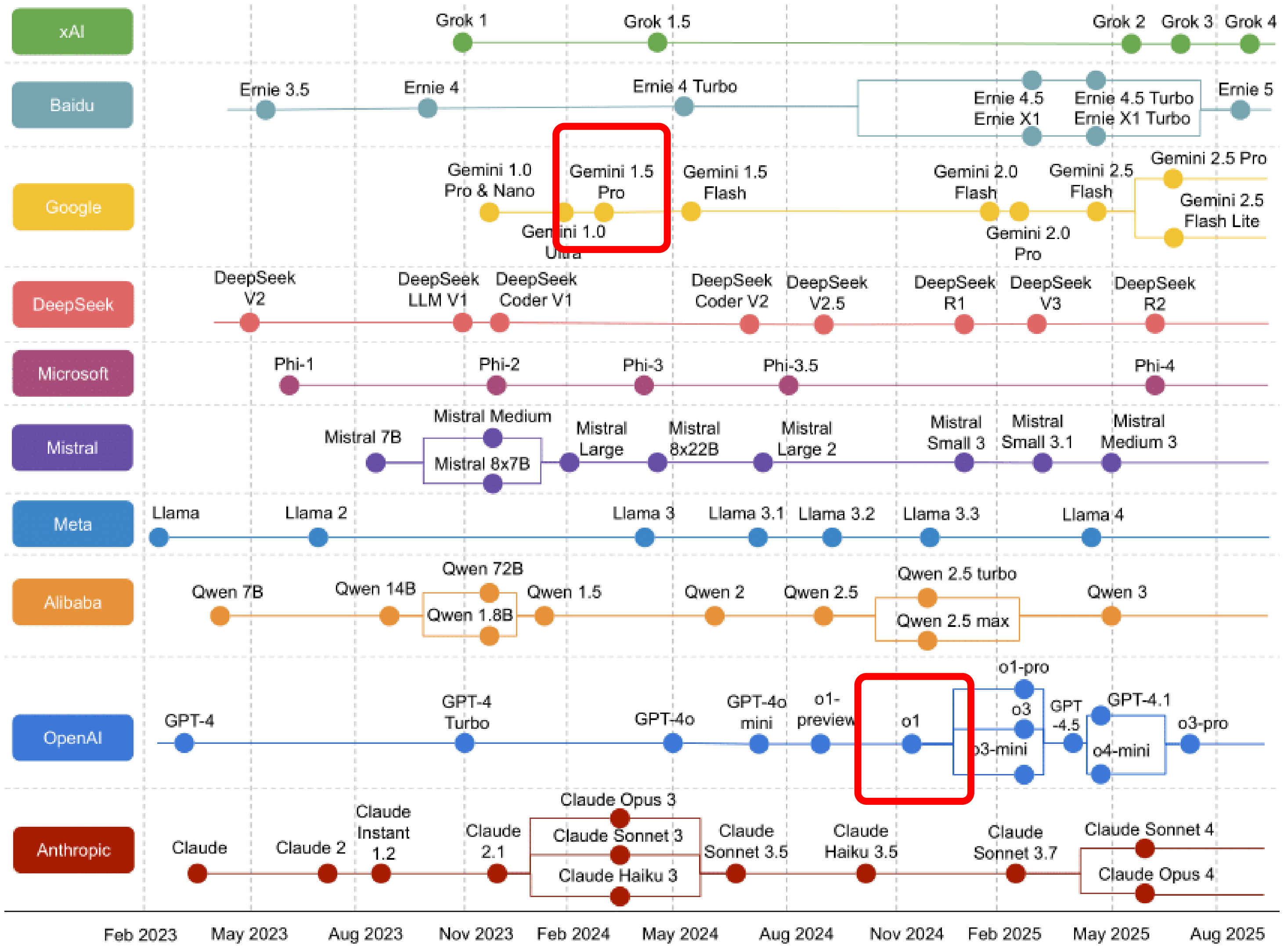


https://medium.com/@tenyks_blogger/multimodal-large-language-models-mllms-transforming-computer-vision-76d3c5dd267f

Flamingo: a Visual Language Model for Few-Shot Learning



Reasoning Models (2024-2025)



1M token context window (hour-long video)

Chain-of-Thought (CoT) reasoning, solving 83% of IMO questions

https://arxiv.org/abs/2507.18479

As of 3 March 2026

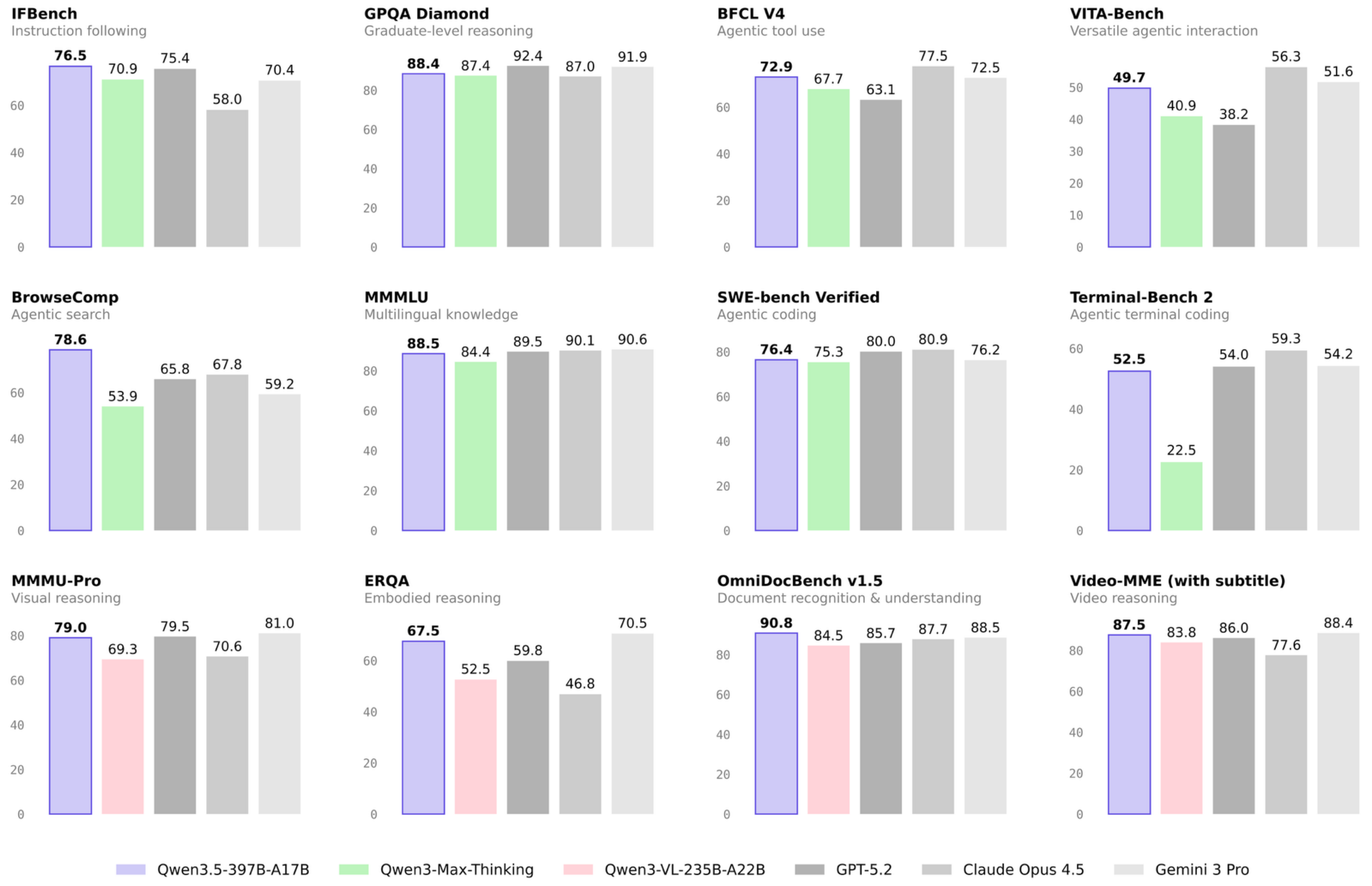
Closed-source (proprietary flagship)

- Gemini 3.1 Pro (Google)
- GPT 5.2 (OpenAI)
- Claude 4.6 (Anthropic)
- Grok 4.1 (xAI)
- Seed 2.0 (ByteDance)
- ...

Open-weight

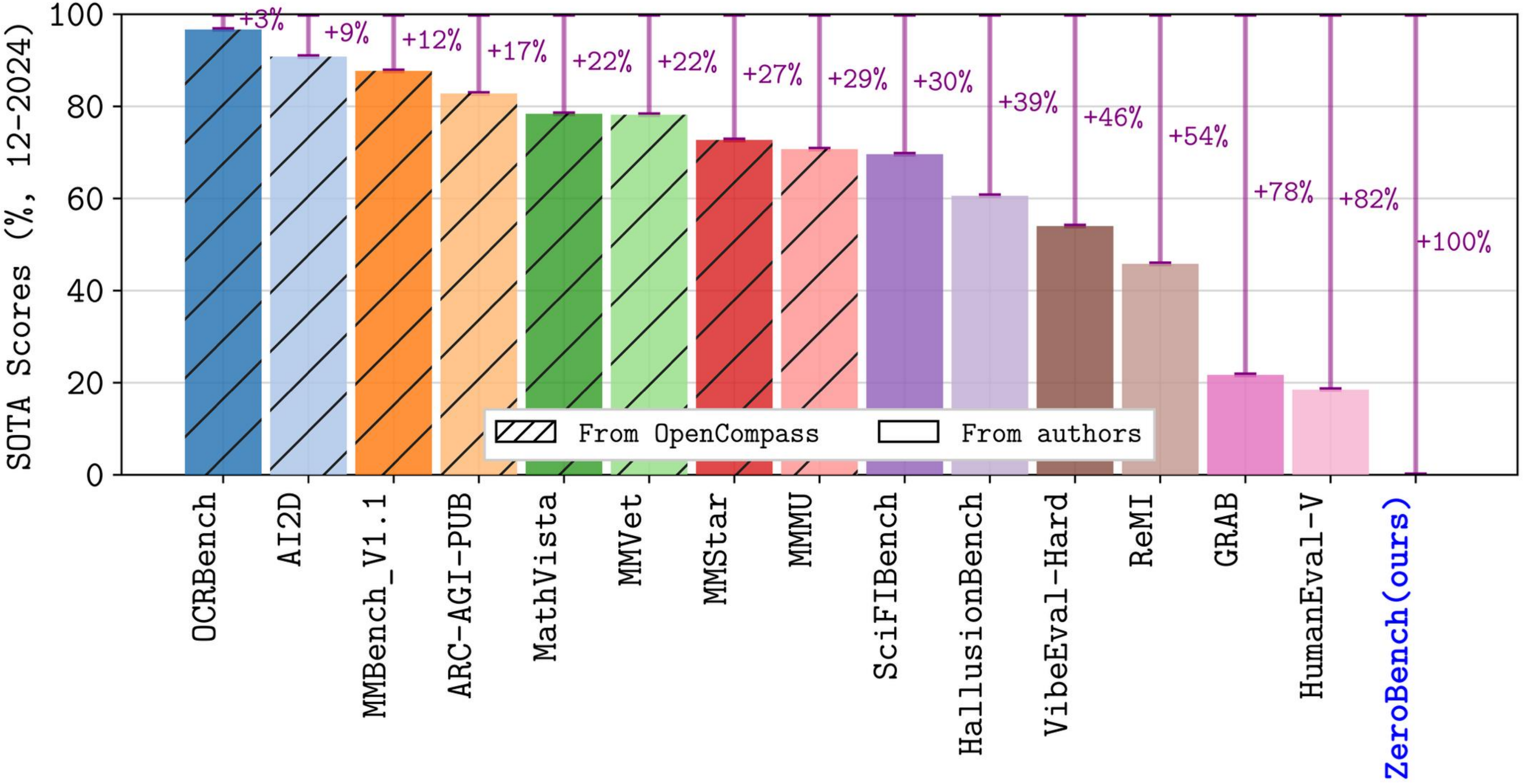
- Qwen 3.5 (Alibaba)
- Kimi K2.5 (Moonshot AI)
- DeepSeek V3.2 (DeepSeek)
- GLM 5 (Zhipu)
- Gemma 3 (Google)
- Mistral 3 (Mistral AI)
- Llama 4 (Meta)
- GPT-OSS-120B (OpenAI)
- ...

Evaluation Benchmarks



ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models



ZeroBench. J. Roberts et al. 2025.

ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models

ZeroBench

Models	**Main questions** (100) k/k [%] (n) k=5	pass@k [%] (n) k=1	pass@k [%] (n) k=5	**Subquestions** (334) pass@k [%] (SE_{CLT}) k=1	Num. correct k=1
Reasoning LMMs					
o1 pro[◇,§]	0.0 (0)	0.0 (0)	-	22.40 (2.48)	75
o1[◇]	0.0 (0)	0.0 (0)	0.0 (0)	20.21 (2.33)	68
Gemini 2 Flash Thinking	0.0 (0)	0.0 (0)	**5.0 (5)**	20.51 (2.60)	69
QVQ	0.0 (0)	0.0 (0)	4.0 (4)	20.47 (2.38)	70
Proprietary LMMs					
GPT-4o	0.0 (0)	0.0 (0)	0.0 (0)	19.60 (2.37)	67
GPT-4o mini	0.0 (0)	0.0 (0)	2.0 (2)	16.58 (2.41)	54
Gemini 2 Flash	0.0 (0)	0.0 (0)	3.0 (3)	23.24 (2.85)	75
Gemini 1.5 Pro	0.0 (0)	0.0 (0)	2.0 (2)	20.88 (2.47)	74
Gemini 1.5 Flash	0.0 (0)	0.0 (0)	2.0 (2)	17.87 (2.41)	63
Gemini 1 Pro Vision	0.0 (0)	0.0 (0)	2.0 (2)	12.36 (2.08)	46
Claude 3.5 Sonnet v2	0.0 (0)	0.0 (0)	2.0 (2)	**25.50 (2.67)**	**82**
Claude 3.5 Sonnet	0.0 (0)	0.0 (0)	1.0 (1)	20.71 (2.48)	72
Claude 3 Opus	0.0 (0)	0.0 (0)	0.0 (0)	15.10 (2.16)	45
Claude 3 Sonnet	0.0 (0)	0.0 (0)	1.0 (1)	16.08 (2.25)	49
Claude 3 Haiku	0.0 (0)	0.0 (0)	0.0 (0)	12.27 (2.05)	41
Reka Edge	0.0 (0)	0.0 (0)	0.0 (0)	3.74 (0.96)	13
Open-weight LMMs					
Llama 3.2 90B	0.0 (0)	0.0 (0)	0.0 (0)	13.26 (1.92)	48
Qwen2-VL-72B-Instruct	0.0 (0)	0.0 (0)	2.0 (2)	13.00 (2.32)	43
NVLM-D-72B	0.0 (0)	0.0 (0)	1.0 (1)	14.91 (2.36)	51
Pixtral-Large	0.0 (0)	0.0 (0)	3.0 (3)	18.68 (2.26)	62

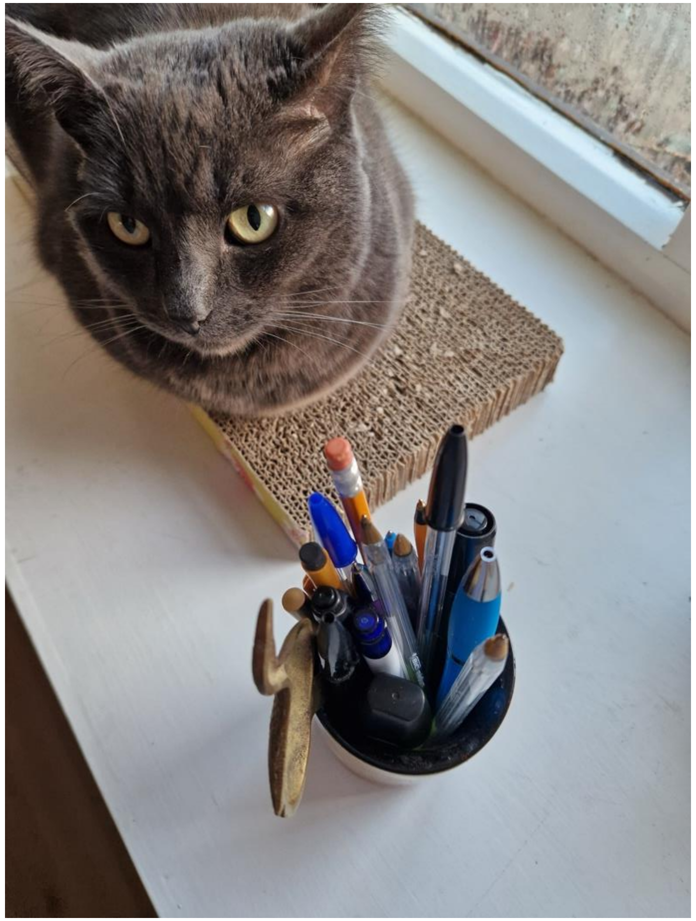
ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models

	Main questions (100)			Subquestions (334)	
Model	pass@1	pass@5	pass^5	pass@1	n correct
Gemini 3.1 Pro 🔥	-	**19.0**	**7.0**	-	-
Gemini 3 Pro 🔥	-	**19.0**	5.0	-	-
GPT-5.2 (medium reasoning) 🔥	-	17.0	6.0	-	-
Gemini 3 Flash 🔥	-	13.0	2.0	-	-
Claude Opus 4.5 🔥	-	10.0	1.0	-	-
o4-mini 🔥	2.0	10.0	0.0	29.05	89
GPT-5-mini (high reasoning) 🔥	4.0	9.0	3.0	27.79	**95**
Claude Opus 4.1 (thinking) 🔥	5.0	8.0	1.0	24.38	80
GPT-5 (medium reasoning) 🔥	1.0	7.0	0.0	26.19	93
GPT 4.5 🔥	1.0	7.0	0.0	27.01	**95**
Claude Sonnet 4 (thinking) 🔥	3.0	6.0	0.0	24.60	80
o3 🔥	3.0	6.0	0.0	25.51	85
GPT-5.1 (medium reasoning) 🔥	-	5.0	0.0	-	-
Claude Opus 4 (thinking) 🔥	4.0	5.0	1.0	25.06	81
Claude Sonnet 4 🔥	2.0	5.0	1.0	23.01	77
Gemini 2.5 Pro 🔥	3.0	5.0	1.0	26.04	90
Gemini 2 Flash Thinking	0.0	5.0	0.0	20.51	69
Claude Opus 4.1 🔥	1.0	4.0	1.0	25.30	81
Grok 4 🔥	1.0	4.0	0.0	21.58	71

ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models



Question:

In the image there is a stationary pot (containing different pens and a letter opener) and other things. Consider: (1) The typical number of legs of the species visible in the picture (2) The total number of pens (3) The total number of pen nibs exposed and visible. What is the value of the product of (1), (2) and (3)?

ZeroBench

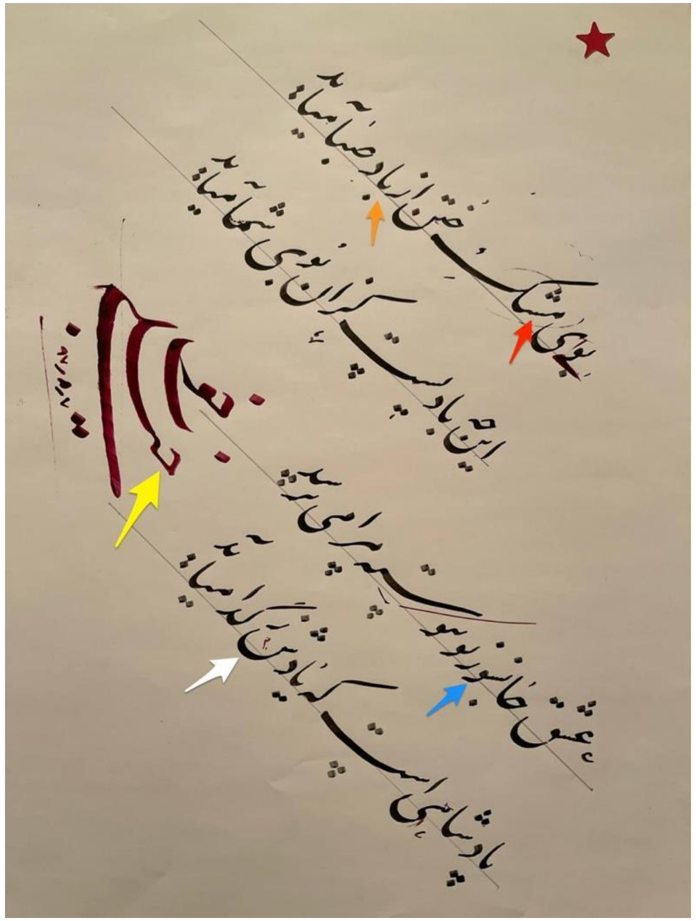
An Impossible* Visual Benchmark for Contemporary Large Multimodal Models

Question:

Read the note pointed at by the pink arrow. Then, read the note pointed at by the green arrow. Return the two notes in that order without a space or comma.

ZeroBench

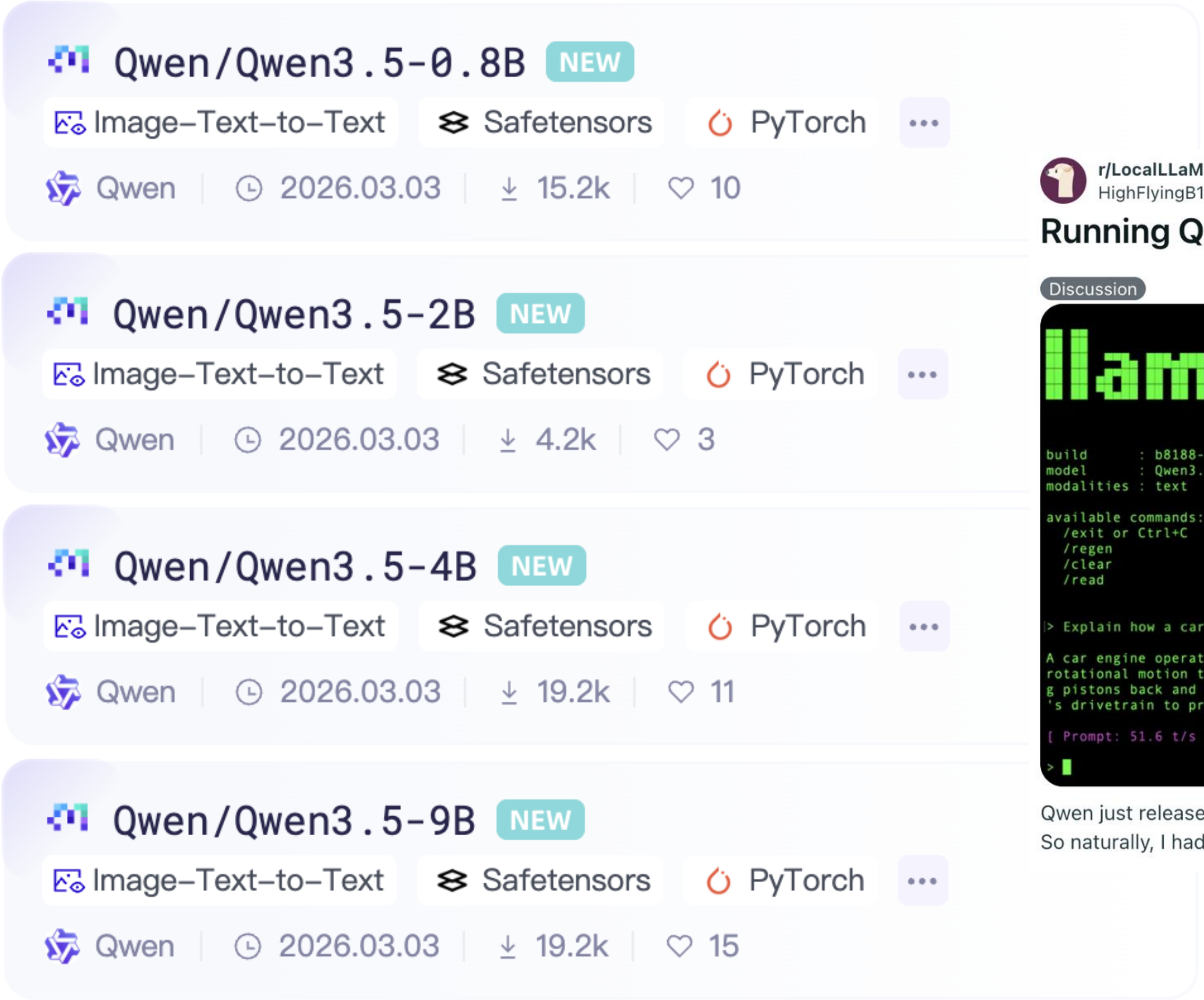
An Impossible* Visual Benchmark for Contemporary Large Multimodal Models



Question:

The image shows calligraphic writing in Farsi, with some letters annotated with arrows. If we put these letters together, what would the message be when translated into English? Use the colors in the following order: red, orange, yellow, blue, and white.

Qwen 3.5 Small Models (3 March 2026)



https://www.reddit.com/r/LocalLLaMA/comments/1rizodv/running_qwen_35_08b_locally_in_the_browser_on/

Qwen 3.5 Small Models

Qwen/Qwen3.5-0.8B NEW
Image-Text-to-Text · Safetensors · PyTorch · ...
Qwen | 2026.03.03 | 15.2k | 10

Qwen/Qwen3.5-2B NEW
Image-Text-to-Text · Safetensors · PyTorch · ...
Qwen | 2026.03.03 | 4.2k | 3

Qwen/Qwen3.5-4B NEW
Image-Text-to-Text · Safetensors · PyTorch · ...
Qwen | 2026.03.03 | 19.2k | 11

Qwen/Qwen3.5-9B NEW
Image-Text-to-Text · Safetensors · PyTorch · ...
Qwen | 2026.03.03 | 19.2k | 15

Fine-tune Qwen3.5

Train Qwen3.5 locally on just 5GB VRAM.

You can now fine-tune Qwen3.5 model family (0.8B, 2B, 4B, 9B, 27B, 35B-A3B, 122B-A10B) with Unsloth ↗. Support includes vision and text. **Qwen3.5-35B-A3B** - bf16 LoRA works on **74GB VRAM.**

- Unsloth makes Qwen3.5 train **1.5× faster** and uses **50% less VRAM** than FA2 setups.
- Qwen3.5 BF16 LoRA VRAM requirements: **0.8B**: 3GB • **2B**: 5GB • **4B**: 10GB • **27B**: 56GB
- Fine-tune **0.8B, 2B, 4B** and **9B** via our **free Google Colab notebooks**:

Qwen3.5-**0.8B** ↗ | Qwen3.5-**2B** ↗ | Qwen3.5-**4B** ↗ | unsloth

- If you want to **preserve reasoning** ability, you can mix reasoning-style examples with direct answers (keep a minimum of 75% reasoning). Otherwise you can emit it fully.
- After fine-tuning, you can export to GGUF (for llama.cpp/Ollama/LM Studio/etc.) or vLLM
- Reinforcement Learning (RL) for Qwen3.5 VLM RL also works via Unsloth inference.
- We have **A100** Colab notebooks for Qwen3.5-27B ↗ and Qwen3.5-35B-A3B ↗.

If you're on an older version (or fine-tuning locally), update first:

```
pip install --upgrade --force-reinstall --no-cache-dir unsloth unsloth_zoo
```

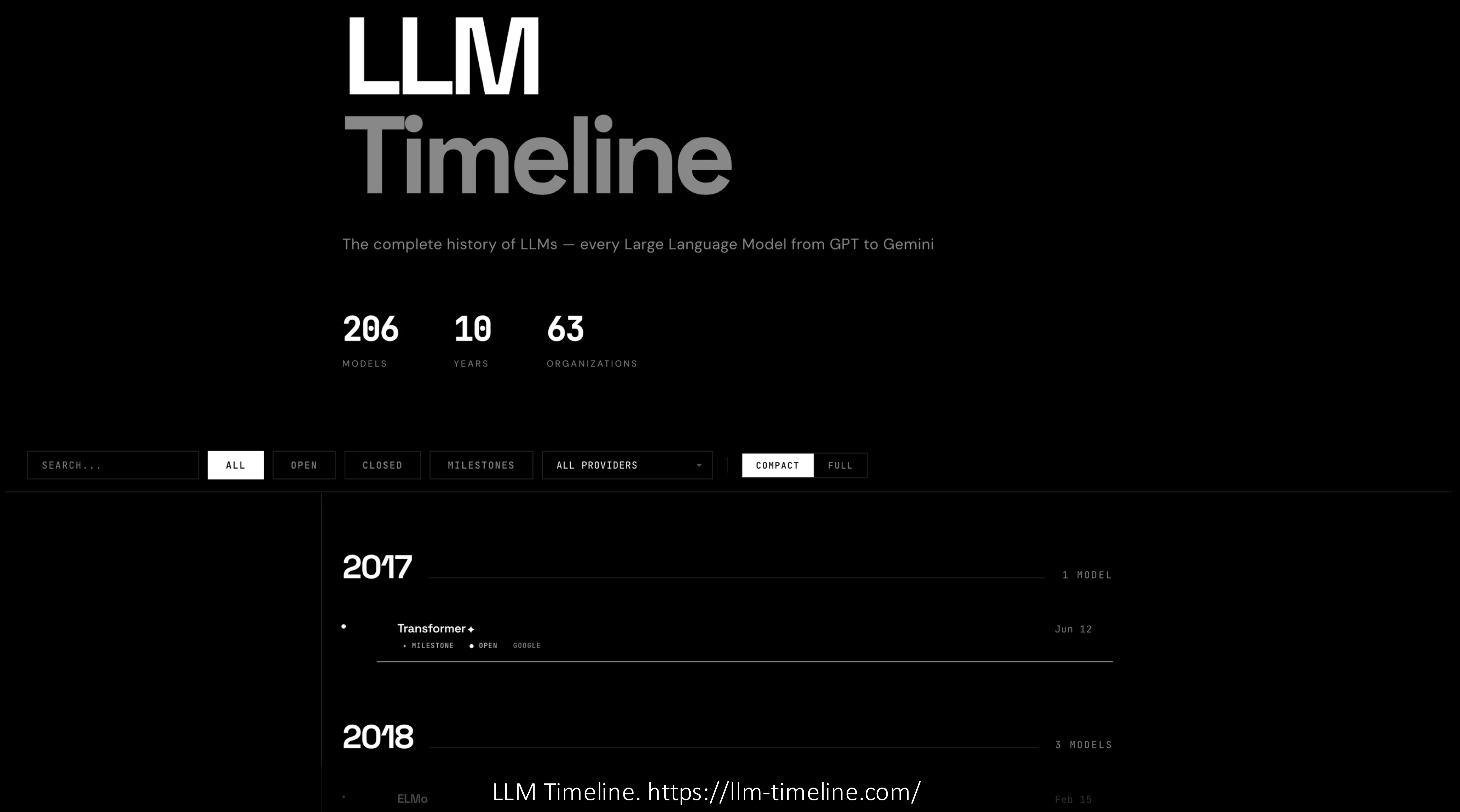
If training seems **slower than usual**, it's because Qwen3.5 use custom Mamba Triton kernels. Compiling those kernels can take longer than normal, especially on T4 GPUs.

Below is a minimal SFT recipe (works for "text-only" fine-tuning). See also vision fine-tuning section.

```
from unsloth import FastLanguageModel
import torch
from datasets import load_dataset
from trl import SFTTrainer, SFTConfig

max_seq_length = 2048  # start small; scale up after it works

# Example dataset (replace with yours). Needs a "text" column.
url = "https://huggingface.co/datasets/laion/OIG/resolve/main/unified_chip2.jsonl"
dataset = load_dataset("json", data_files={"train": url}, split="train")
```



LLM Timeline. https://llm-timeline.com/

In-context Learning

Language Models are Few-Shot Learners

OpenAI

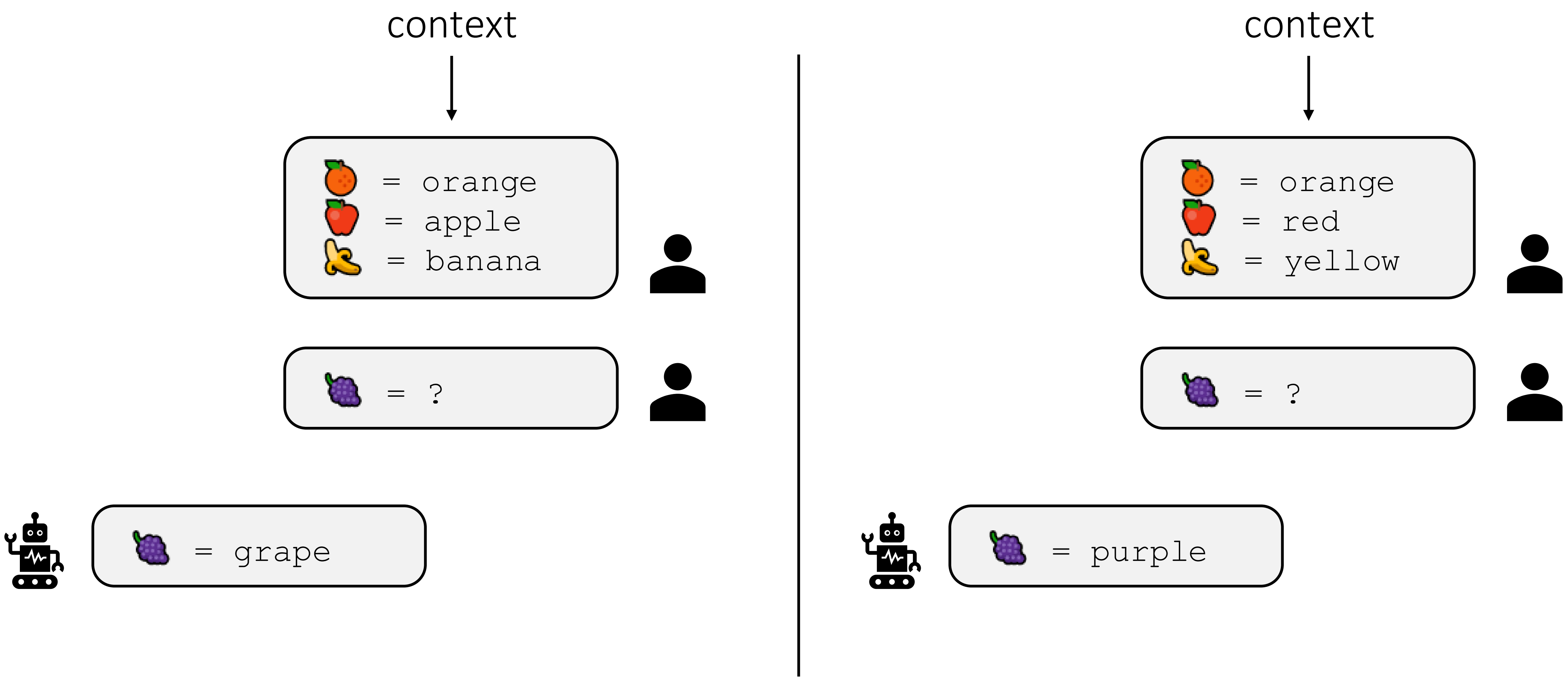
GPT-3 Paper

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

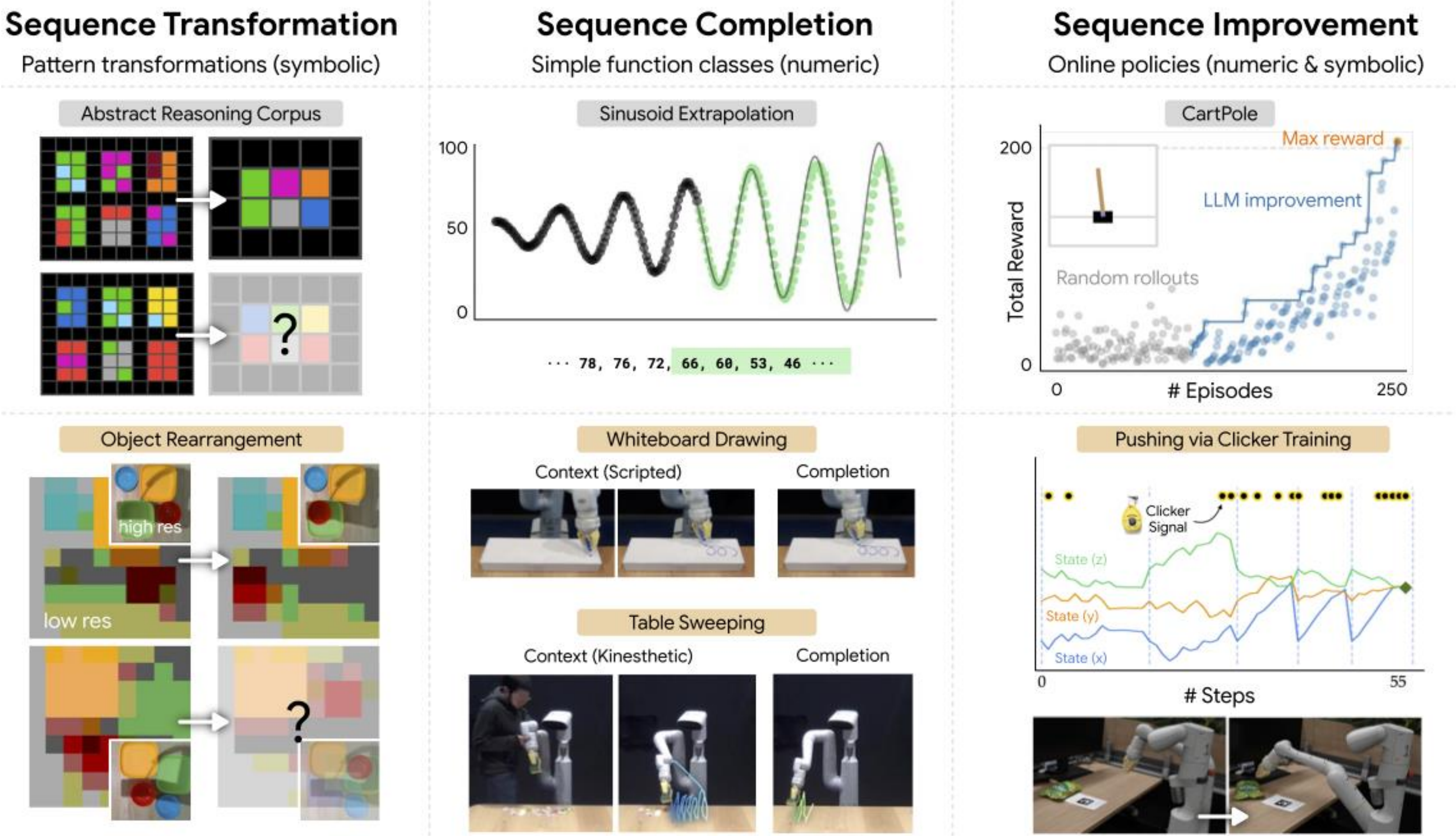
GPT-3. OpenAI. NeurIPS 2020.

In-context Learning



Large Language Models as General Pattern Machines

Suvir Mirchandani Fei Xia Pete Florence Brian Ichter Danny Driess Montserrat Gonzalez Arenas
Kanishka Rao Dorsa Sadigh Andy Zeng
Google DeepMind



An Explanation of In-context Learning as Implicit Bayesian Inference

Sang Michael Xie
Stanford University
xie@cs.stanford.edu

Aditi Raghunathan
Stanford University
aditir@stanford.edu

Percy Liang
Stanford University
pliang@cs.stanford.edu

Tengyu Ma
Stanford University
tengyuma@cs.stanford.edu

Abstract

Large language models (LMs) such as GPT-3 have the surprising ability to do in-context learning, where the model learns to do a downstream task simply by conditioning on a prompt consisting of input-output examples. The LM learns from these examples *without being explicitly pretrained to learn*. Thus, it is unclear what enables in-context learning. In this paper, we study how in-context learning can emerge when pretraining documents have long-range coherence. Here, the LM must infer a latent document-level concept to generate coherent next tokens during pretraining. At test time, in-context learning occurs when the LM also infers a shared latent concept between examples in a prompt. We prove when this occurs despite a distribution mismatch between prompts and pretraining data in a setting where the pretraining distribution is a mixture of HMMs. In contrast to messy large-scale datasets used to train LMs capable of in-context learning, we generate a small-scale synthetic dataset (GINC) where Transformers and LSTMs both exhibit in-context learning[1]. Beyond the theory, experiments on GINC exhibit large-scale real-world phenomena including improved in-context performance with model scaling (despite the same pretraining loss), sensitivity to example order, and instances where zero-shot is better than few-shot in-context learning.

S. Xie et al. ICLR 2022.

Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models

Qizheng Zhang [1*] Changran Hu [2*] Shubhangi Upasani [2] Boyuan Ma [2] Fenglu Hong [2]
Vamsidhar Kamanuru [2] Jay Rainton [2] Chen Wu [2] Mengmeng Ji [2] Hanchen Li [3]
Urmish Thakker [2] James Zou [1] Kunle Olukotun [1]

[1] Stanford University [2] SambaNova Systems, Inc. [3] UC Berkeley * equal contribution

qizhengz@cs.stanford.edu, changran_hu@berkeley.edu, kunle@cs.stanford.edu
https://github.com/ace-agent/ace https://ace-agent.github.io

Abstract

Large language model (LLM) applications such as agents and domain-specific reasoning increasingly rely on *context adaptation*—modifying inputs with instructions, strategies, or evidence, rather than weight updates. Prior approaches improve usability but often suffer from brevity bias, which drops domain insights for concise summaries, and from context collapse, where iterative rewriting erodes details over time. Building on the adaptive memory introduced by Dynamic Cheatsheet, we introduce ACE (**A**gentic **C**ontext **E**ngineering), a framework that treats contexts as evolving playbooks that accumulate, refine, and organize strategies through a modular process of generation, reflection, and curation. ACE prevents collapse with structured, incremental updates that preserve detailed knowledge and scale with long-context models. Across agent and domain-specific benchmarks, ACE optimizes contexts both offline (*e.g.*, system prompts) and online (*e.g.*, agent memory), consistently outperforming strong baselines: +10.6% on agents and +8.6% on finance, while significantly reducing adaptation latency and rollout cost. Notably, ACE could adapt effectively without labeled supervision and instead by leveraging natural execution feedback. On the AppWorld leaderboard, ACE matches the top-ranked production-level agent on the overall average and surpasses it on the harder test-challenge split, despite using a smaller open-source model. These results show that comprehensive, evolving contexts enable scalable, efficient, and self-improving LLM systems with low overhead.

Context engineering as opposed to fine-tuning in the age of large models?

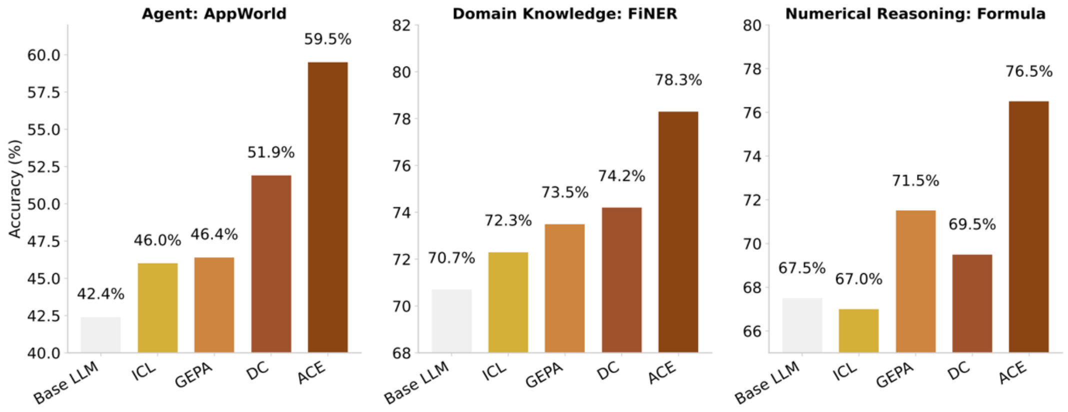


Figure 1: **Overall Performance Results.** Our proposed framework, ACE, consistently outperforms strong baselines across agent and domain-specific reasoning tasks.

Visual Generation

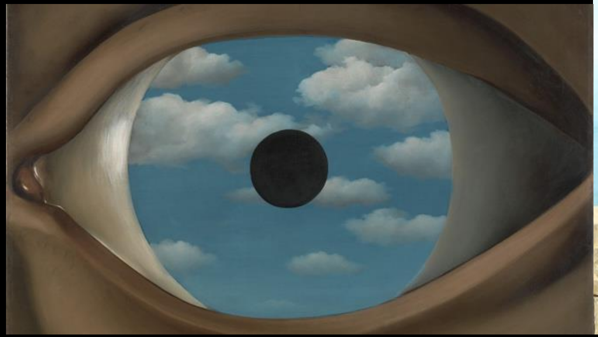
DALL·E 2. OpenAI. 2022.

Demonstrated the power of diffusion models

"An IT-guy trying to fix hardware of a PC tower is being tangled by the PC cables like Laokoon. Marble, copy after Hellenistic original from ca. 200 BC. Found in the Baths of Trajan, 1506."

Generated by DALL·E 2

"About 99% of the time, the right time is right now in René Magritte style."



"The False Mirror" (1928)

"The Son of Man" (1964)

Generated by DALL·E 2

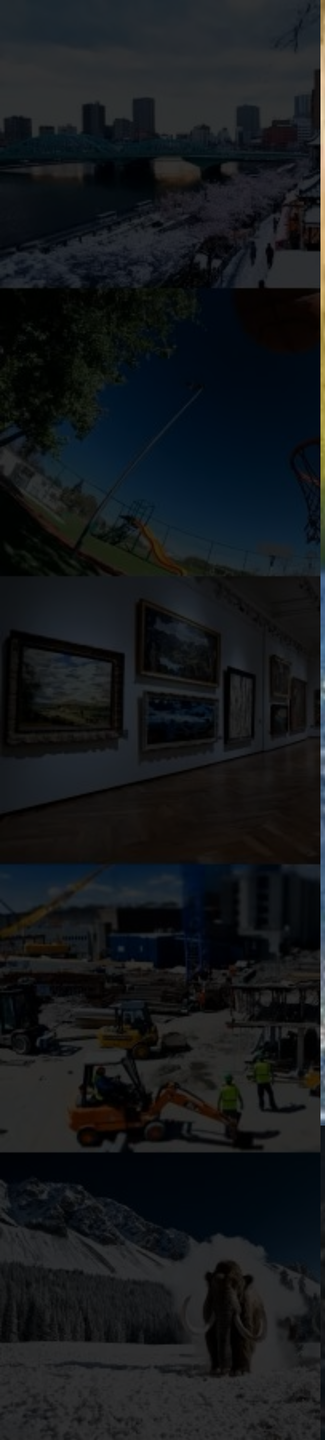
Sora. OpenAI. 2024.

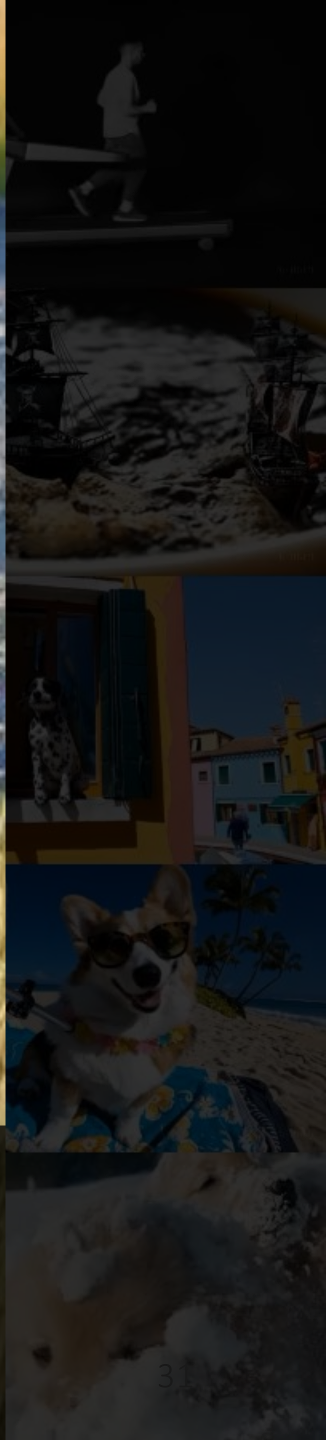
Demonstrated the power of Diffusion Transformer (DiT)

Sora: Video Generation Models as World Simulators. OpenAI. 2024.

Pixels Defying Physics

Sora: Video Generation Models as World Simulators. OpenAI. 2024.





Pixels Defying Physics

Sora: Video Generation Models as World Simulators. OpenAI. 2024.

Infinite Interactive Worlds

Project Genie. Google DeepMind. 2026.

Reality Check

First frame from Physics IQ Benchmark, video generated by Veo 3.1 from Google DeepMind with the prompt: “what will happen if the spinning wheel keeps turning and the stick hits the first block on the right.” Adapted from Ayush Tewari.

WorldScore

A Unified Evaluation Benchmark for World Generation

Models	WorldScore-Static	WorldScore-Dynamic	Camera Ctrl	Object Ctrl	Content Align	3D Consist	Photo Consist	Style Consist	Subjective Qual	Motion Acc	Motion Mag	Motion Smooth
Gen-3	60.71	57.58	29.47	62.92	50.49	68.31	87.09	62.82	63.85	54.53	27.48	68.87
Hailuo	57.55	56.36	22.39	**69.56**	73.53	67.18	62.82	54.91	52.44	63.46	27.20	70.07
DynamiCrafter	52.09	47.19	25.15	47.36	25.00	72.90	60.95	78.85	54.40	41.11	39.25	26.92
VideoCrafter1-T2V	47.10	43.54	21.61	50.44	60.78	64.86	51.36	38.05	42.63	11.76	**75.00**	18.87
VideoCrafter1-I2V	50.47	47.64	25.46	24.25	35.27	74.42	73.89	65.17	54.85	55.63	25.00	42.49
VideoCrafter2	52.57	47.49	28.92	39.07	72.46	65.14	61.85	43.79	56.74	47.12	30.40	29.39
T2V-Turbo	45.65	40.20	27.80	30.68	69.14	38.72	34.84	49.65	68.74	34.87	40.09	7.48
EasyAnimate	52.85	51.65	26.72	54.50	50.76	67.29	47.35	73.05	50.31	75.00	31.16	40.32
CogVideoX-T2V	54.18	48.79	40.22	51.05	68.12	68.81	64.20	42.19	44.67	25.00	47.31	36.28
CogVideoX-I2V	62.15	**59.12**	38.27	40.07	36.73	86.21	88.12	83.22	62.44	69.56	26.42	60.15
Allegro	55.31	51.97	24.84	57.47	51.48	70.50	69.89	65.60	47.41	54.39	40.28	37.81
Vchitect-2.0	42.28	38.47	26.55	49.54	65.75	41.53	42.30	25.69	44.58	33.59	33.81	21.31
SceneScape	50.73	35.51	84.99	47.44	28.64	76.54	62.88	21.85	32.75	0.00	0.00	0.00
Text2Room	62.10	43.47	**94.01**	38.93	50.79	88.71	88.36	37.23	36.69	0.00	0.00	0.00
LucidDreamer	70.40	49.28	88.93	41.18	**75.00**	**90.37**	**90.20**	48.10	58.99	0.00	0.00	0.00
WonderJourney	63.75	44.63	84.60	37.10	35.54	80.60	79.03	62.82	66.56	0.00	0.00	0.00
InvisibleStitch	61.12	42.78	93.20	36.51	29.53	88.51	89.19	32.37	58.50	0.00	0.00	0.00
WonderWorld	72.69	50.88	92.98	51.76	71.25	86.87	85.56	70.57	49.81	0.00	0.00	0.00
4D-fy	27.98	32.10	69.92	55.09	0.85	35.47	1.59	32.04	0.89	22.22	22.88	**80.06**
LTX-Video	55.44	56.54	25.06	53.41	39.73	78.41	88.92	53.50	49.08	**76.22**	29.95	71.09
Voyager	**77.62**	54.53	85.95	66.92	68.92	81.56	85.99	**84.89**	**71.09**	0.00	0.00	0.00
Wan2.1	57.56	52.85	23.53	40.32	45.44	78.74	78.36	77.18	59.38	54.27	33.26	38.05

Image Understanding via Conditional Generation

Visual Prompting via Image Inpainting

Amir Bar* [12], Yossi Gandelsman* [1], Trevor Darrell [1], Amir Globerson [2], Alexei A. Efros [1]

UC Berkeley[1] Tel Aviv University[2]

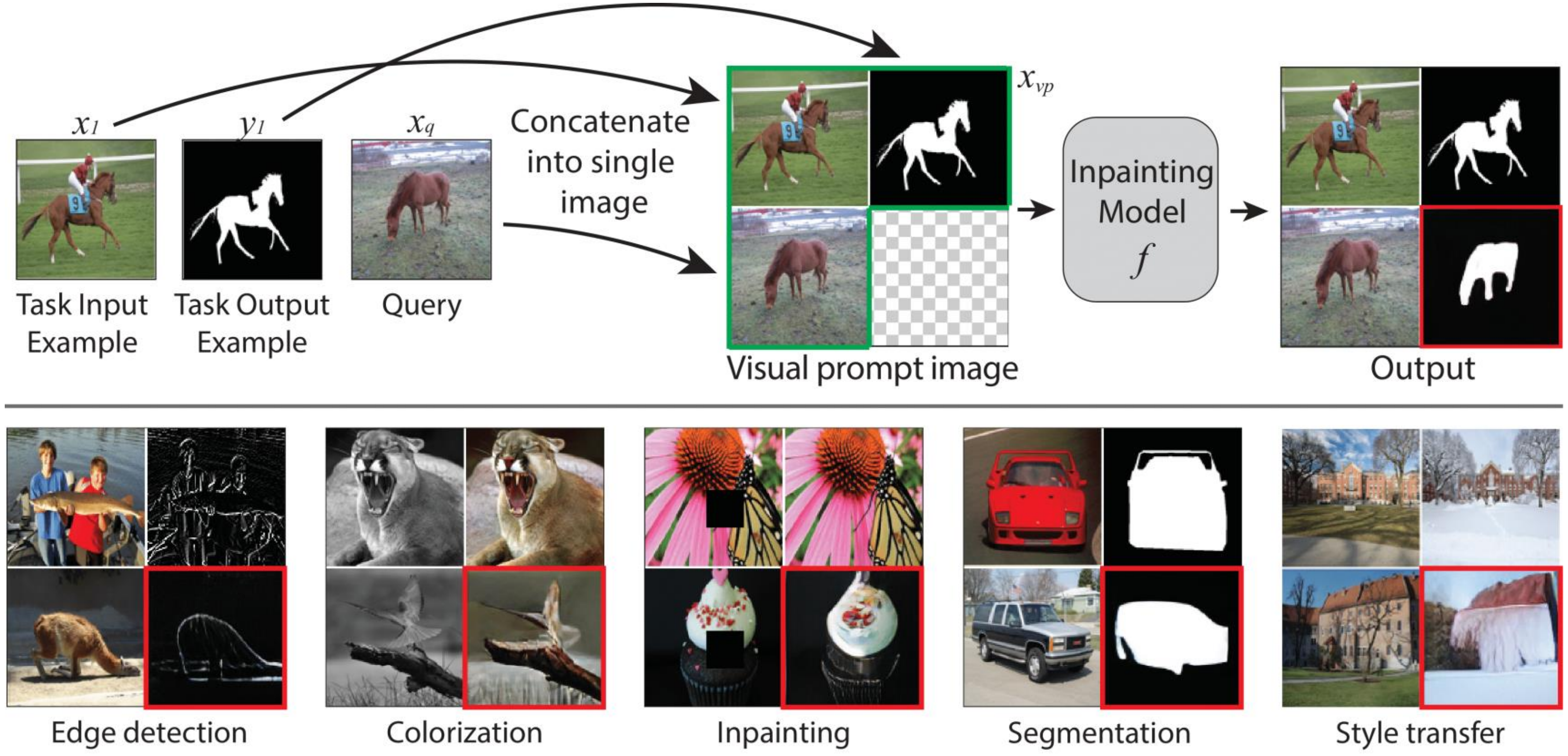


Image Understanding via Conditional Generation

automatic 3D data labelling pipeline using Gemini (Nano Banana 2)

Introducing Meta Segment Anything Model 3 (SAM 3)

SAM 3: Segment Anything with Concepts. Meta. 2025. https://ai.meta.com/research/sam3/

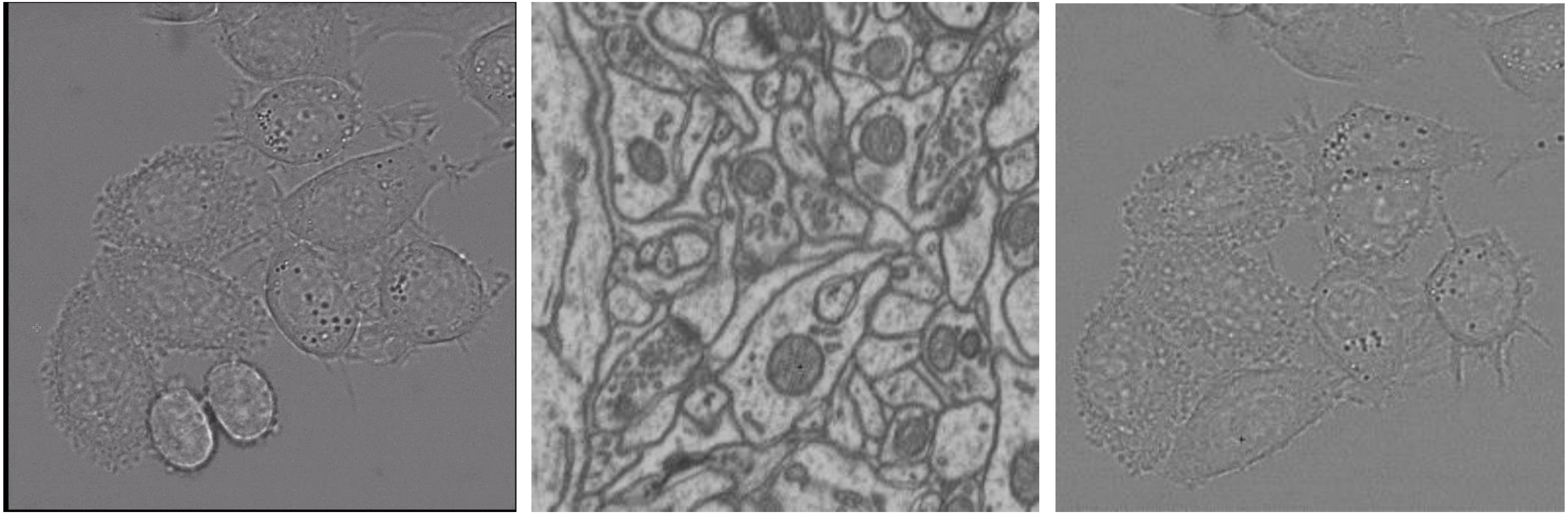
Segment Anything Model 3 (SAM 3)

Meta SAM 3D

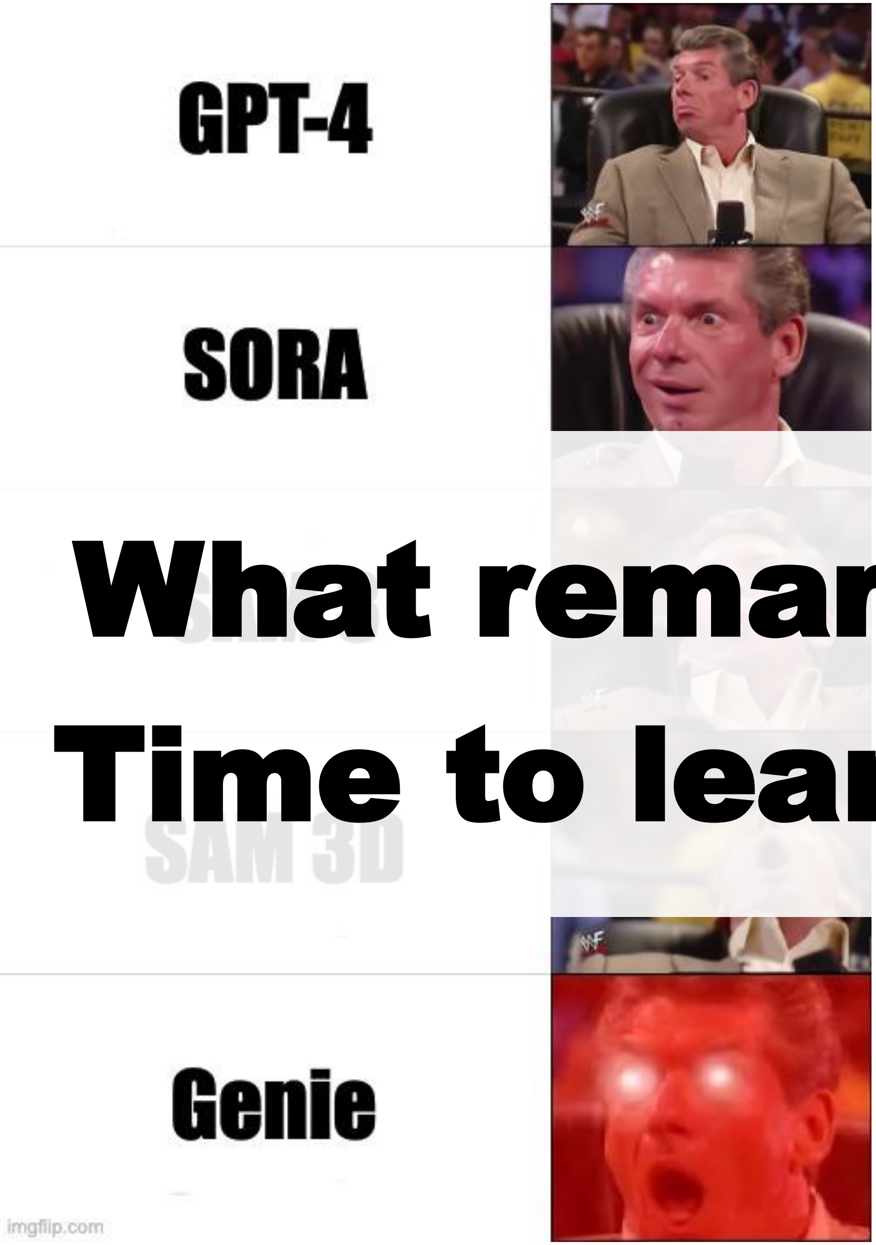
SAM 3D. Meta. 2025. https://ai.meta.com/research/sam3d/

SAM 3D

Segment Anything for Microscopy

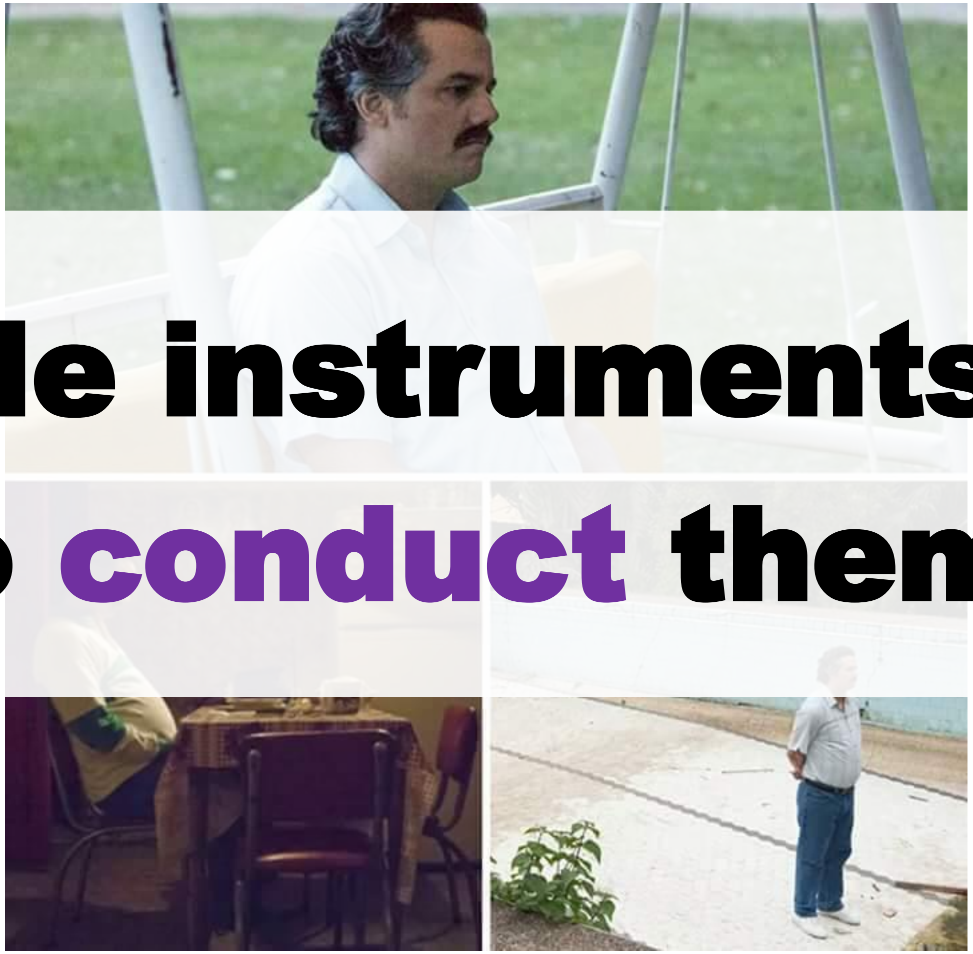


https://github.com/computational-cell-analytics/micro-sam



⋮

Everyone else scaling to 10T parameters...



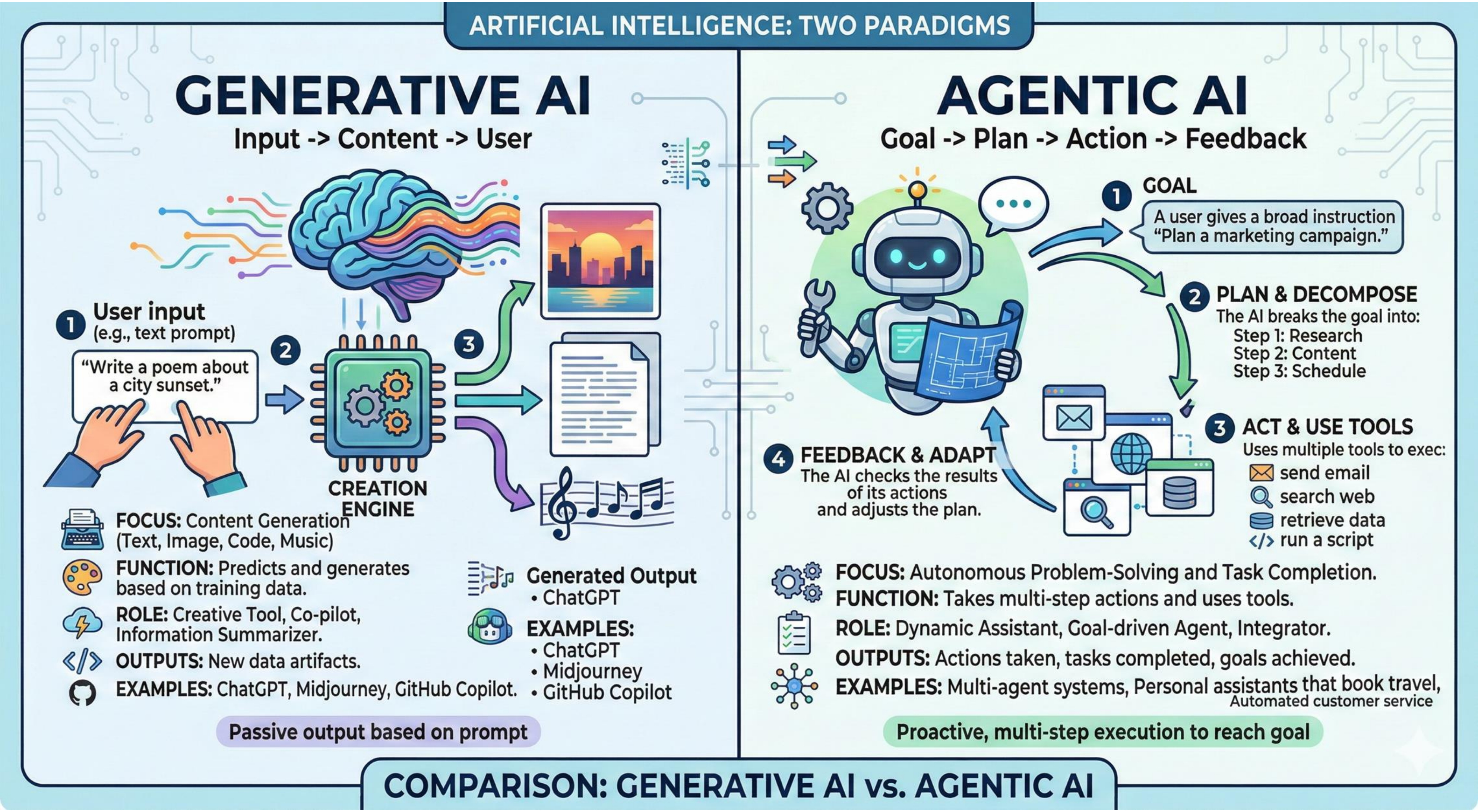
Me with Megalo-Modelo-Phobia.

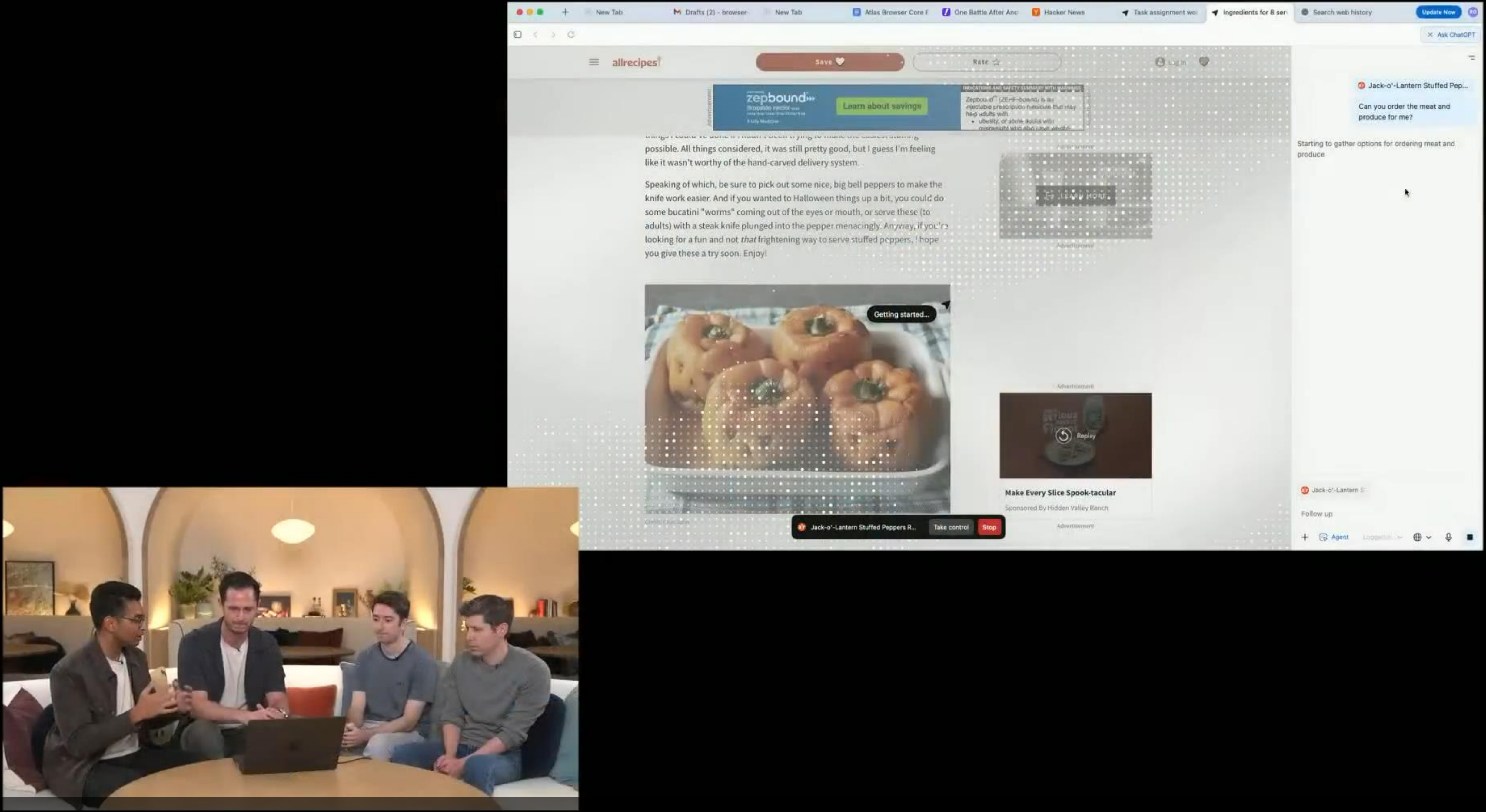
What remarkable instruments!

Time to learn to **conduct** them!

Part 2 – Expanding Horizons

Agentic AI Systems (2024-now)





Introducing ChatGPT Atlas. OpenAI. 2025. https://www.youtube.com/watch?v=8UWKxJbjriY

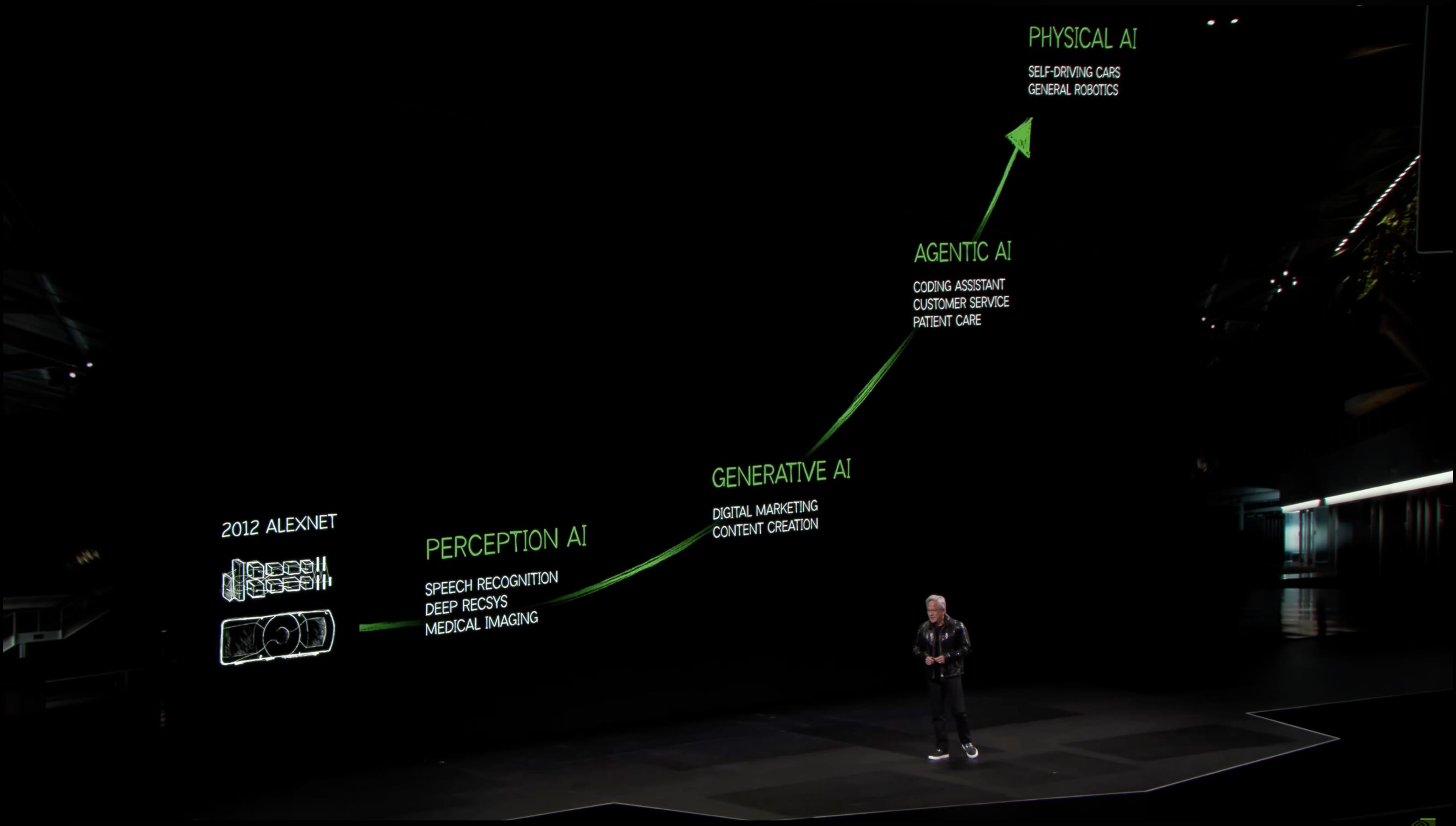
Introducing the Codex app. OpenAI. 2026. https://www.youtube.com/watch?v=HFM3se4lNiw

Introducing Claude Code. Anthropic. 2025. https://www.youtube.com/watch?v=AJpK3YTTKZ4

2.0

Introducing Cursor 2.0. Cursor. 2025. https://www.youtube.com/watch?v=An8IM-kPyms

What’s Next?



NVIDIA CEO Jensen Huang Keynote at CES 2025. https://www.youtube.com/watch?v=k82RwXqZHY8

The Future of Physical AI is Here. NVIDIA. 2026. https://www.youtube.com/watch?v=iWs-2TD5Dcc

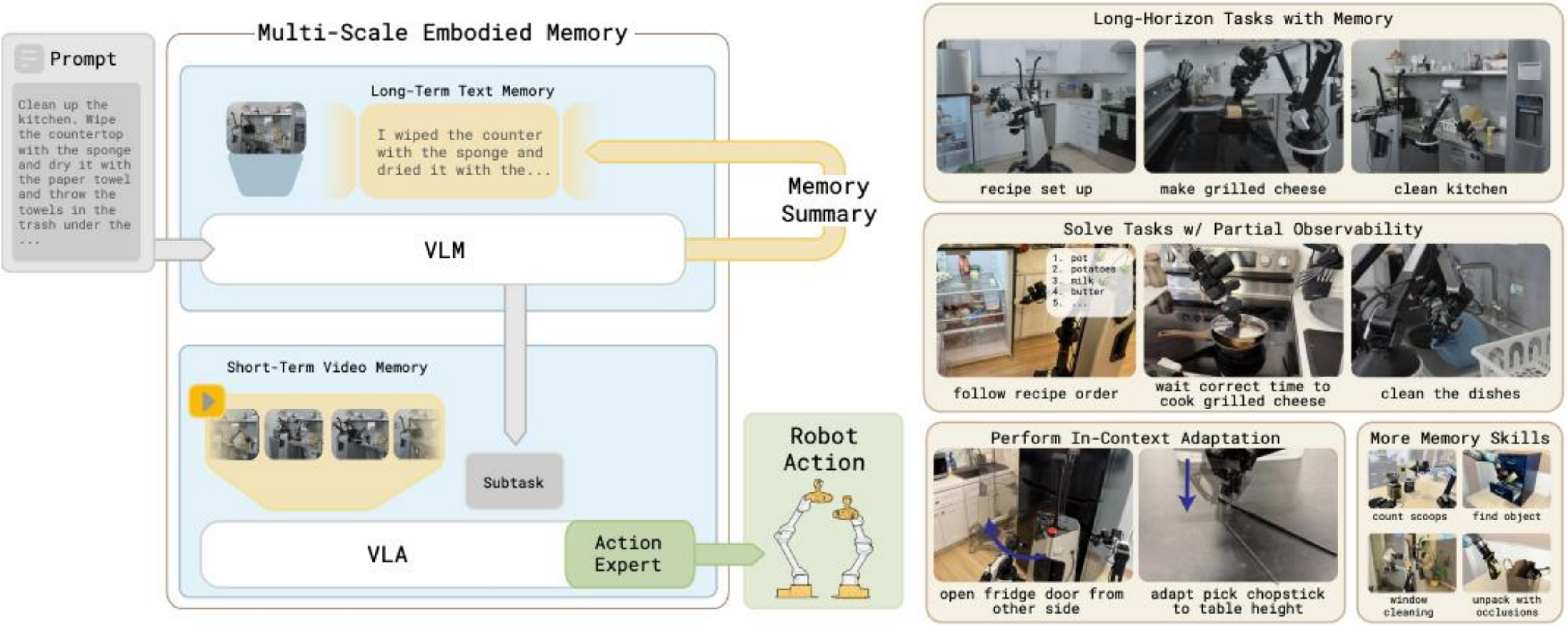
Today's Roadblocks to Physical AI

- Data collection cost
- Physical accuracy (hallucination)
- Memory for long-horizon tasks
- Hardware dexterity
- Latency in deployment
- …

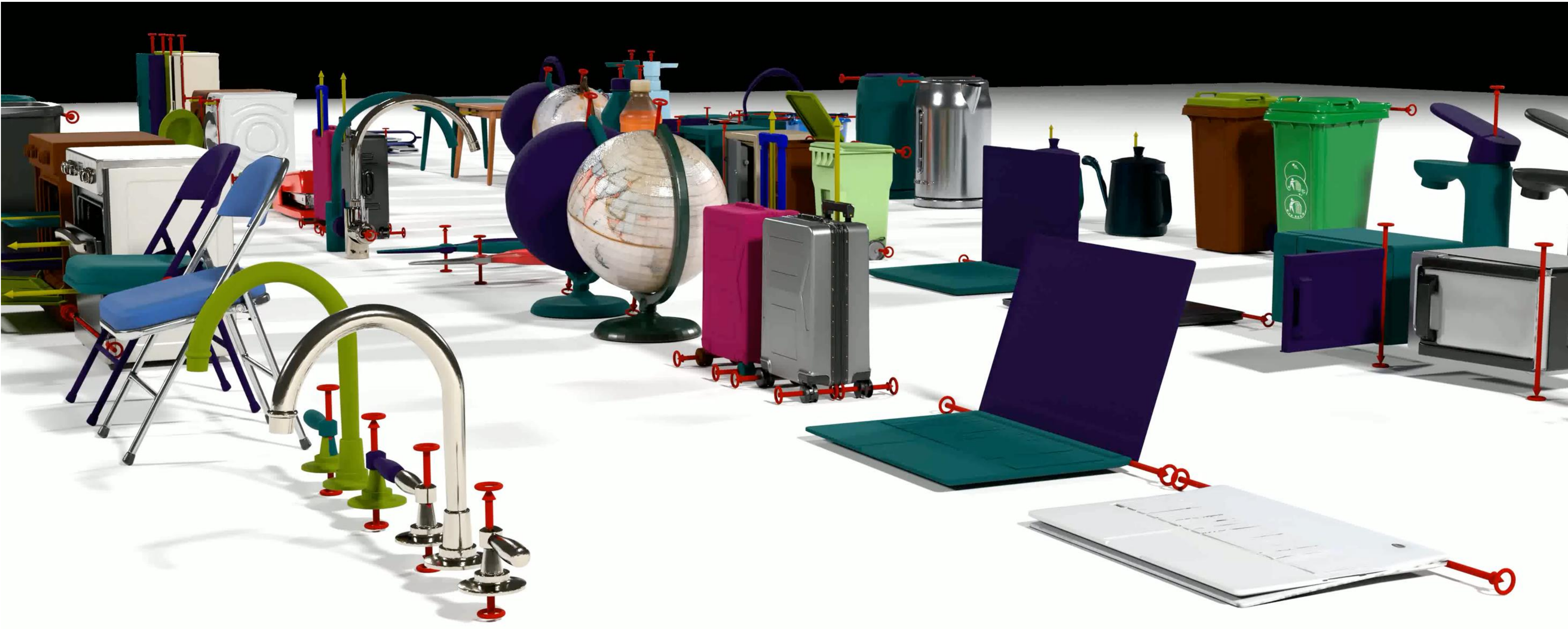
MEM: Multi-Scale Embodied Memory for Vision Language Action Models

Marcel Torne*[1,2]† Karl Pertsch*[1] Homer Walke[3]† Kyle Vedder[1] Suraj Nair[1] Brian Ichter[1]
Allen Z. Ren[1] Haohuan Wang[1] Jiaming Tang[1,4]† Kyle Stachowicz[1,3]† Karan Dhabalia[1] Michael Equi[1]
Quan Vuong[1] Jost Tobias Springenberg[1] Sergey Levine[1] Chelsea Finn[1] Danny Driess[1]

https://pi.website/research/memory



Generative Simulation

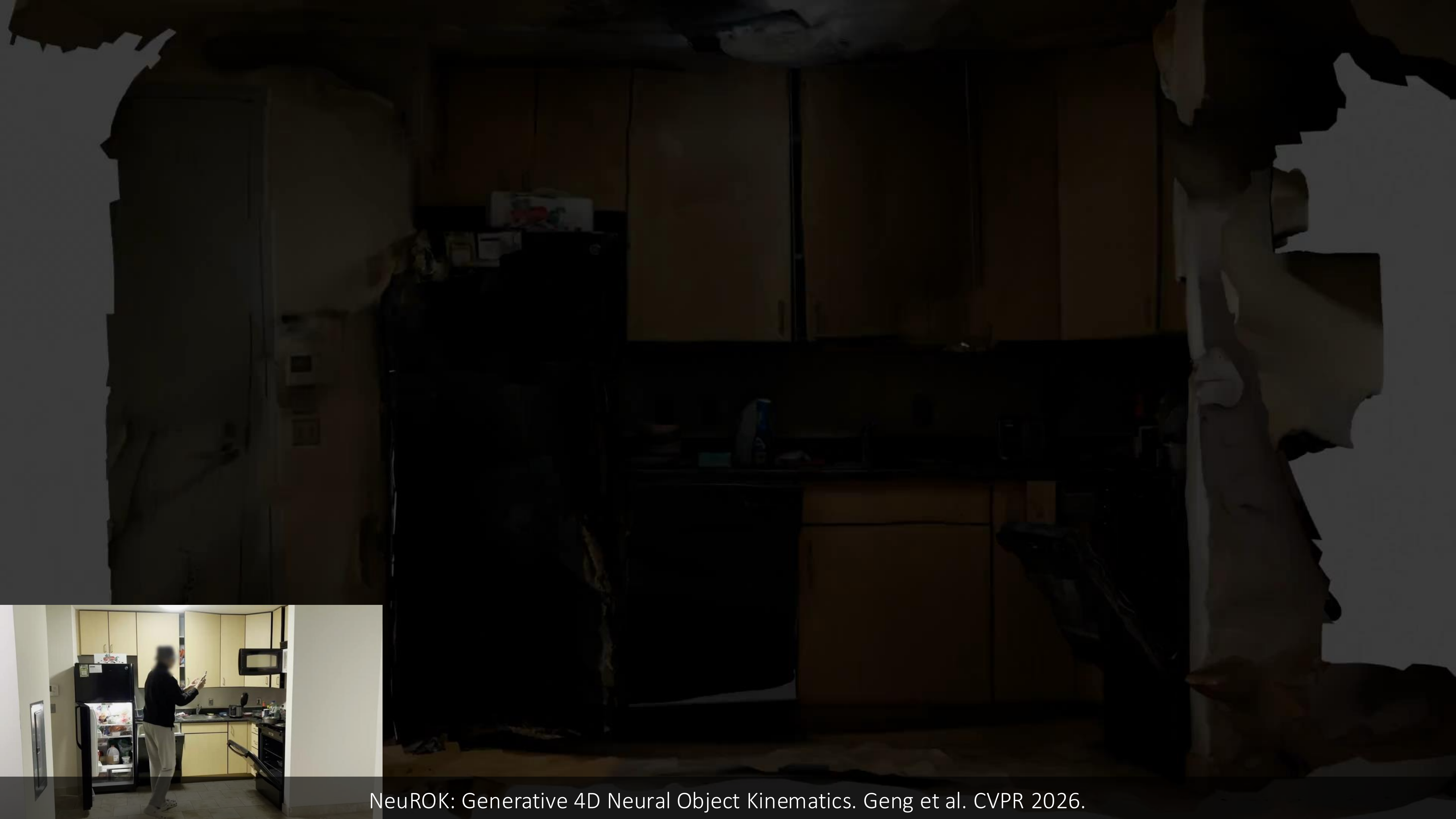


PARTICULATE. Li et al. CVPR 2026.

Generative Simulation

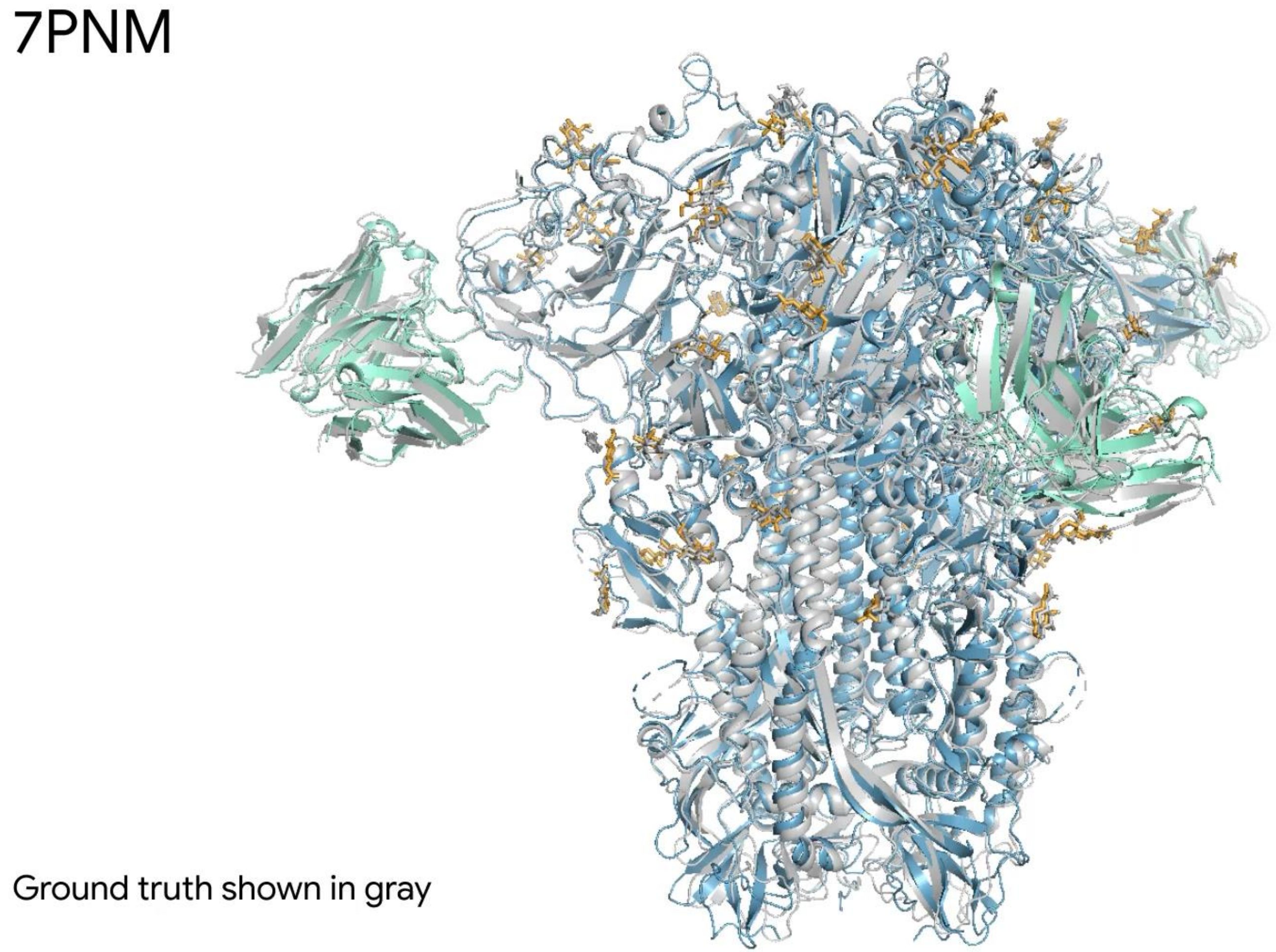
NeuROK: Generative 4D Neural Object Kinematics. Geng et al. CVPR 2026.

NeuROK: Generative 4D Neural Object Kinematics. Geng et al. CVPR 2026.



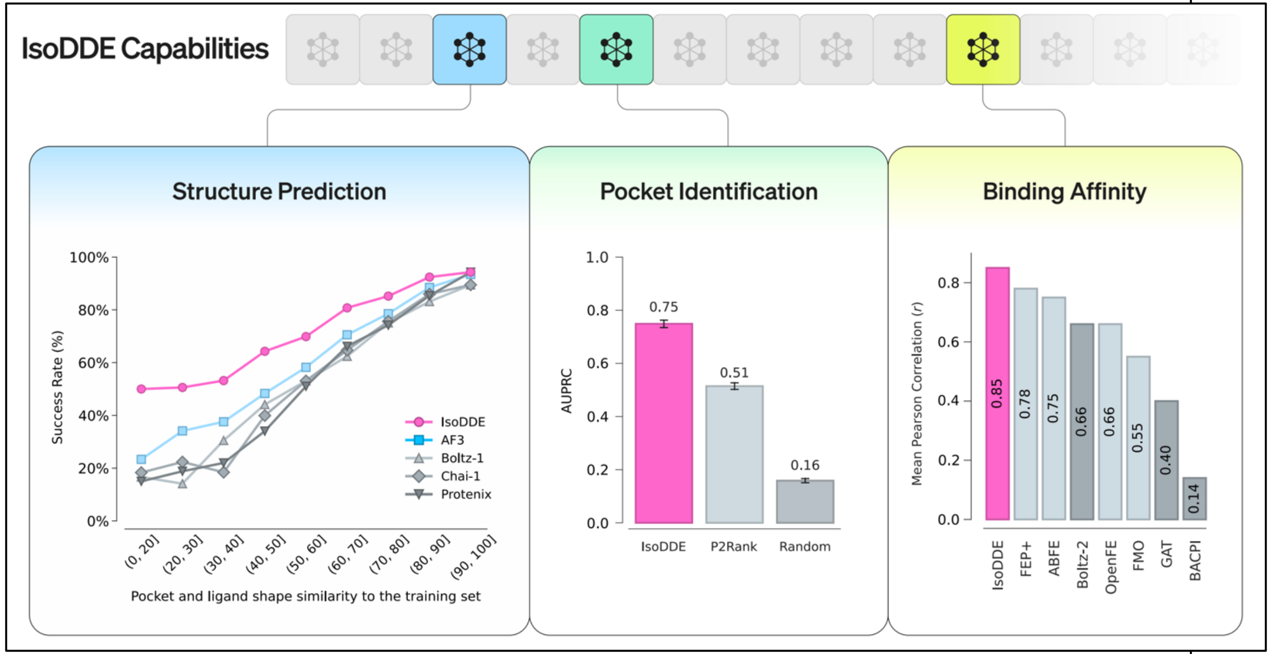
NeuROK: Generative 4D Neural Object Kinematics. Geng et al. CVPR 2026.

AI for Science



AlphaFold 3 – diffusion model predicting 3D folding structure of proteins

AI for Science



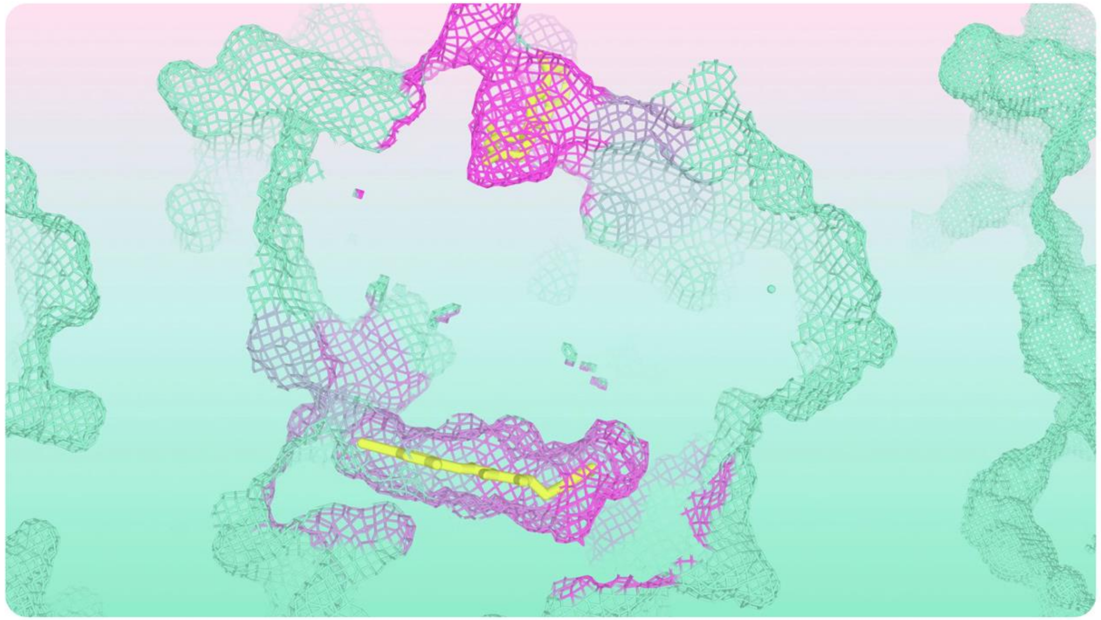
Isomorphic Labs

OUR TEAM | OUR TECH | PARTNERSHIPS | CAREERS | NEWS

VISION

The Isomorphic Labs Drug Design Engine unlocks a new frontier beyond AlphaFold

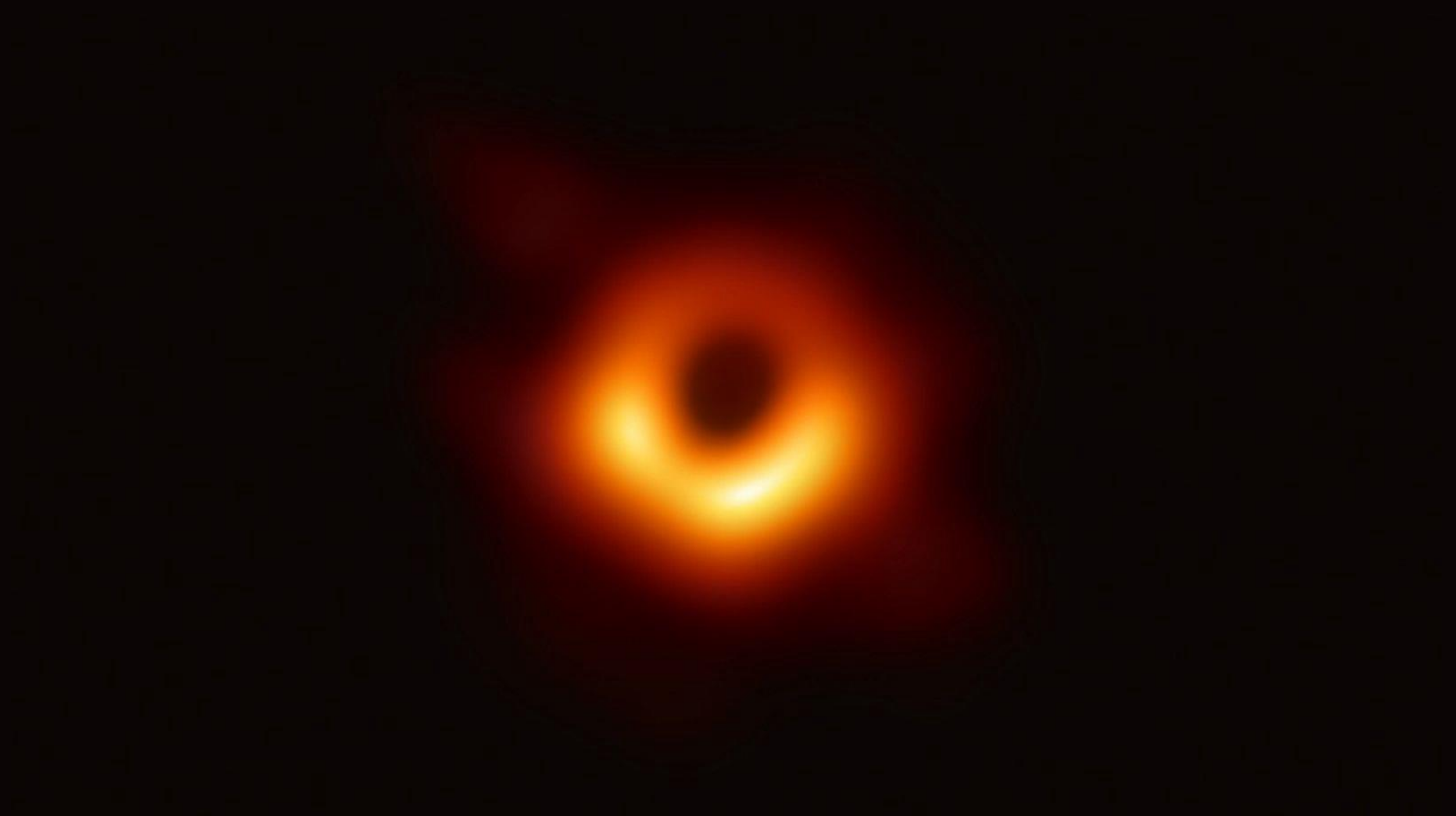
FEBRUARY 10, 2026 | 5 MIN READ | COPY URL



- Structure Prediction of Truly Novel Systems
- Opening a New Window for Complex Biologics
- A New Gold-Standard for Binding Affinity Prediction
- Expanding the Ligandable Proteome
- Advancing Drug Discovery

Today, we are excited to share an update on our progress towards a new frontier of drug design. We have unlocked a new paradigm of predictive accuracy in understanding our biomolecular world, allowing us to rationally design new medicines on a computer with unprecedented understanding and precision.

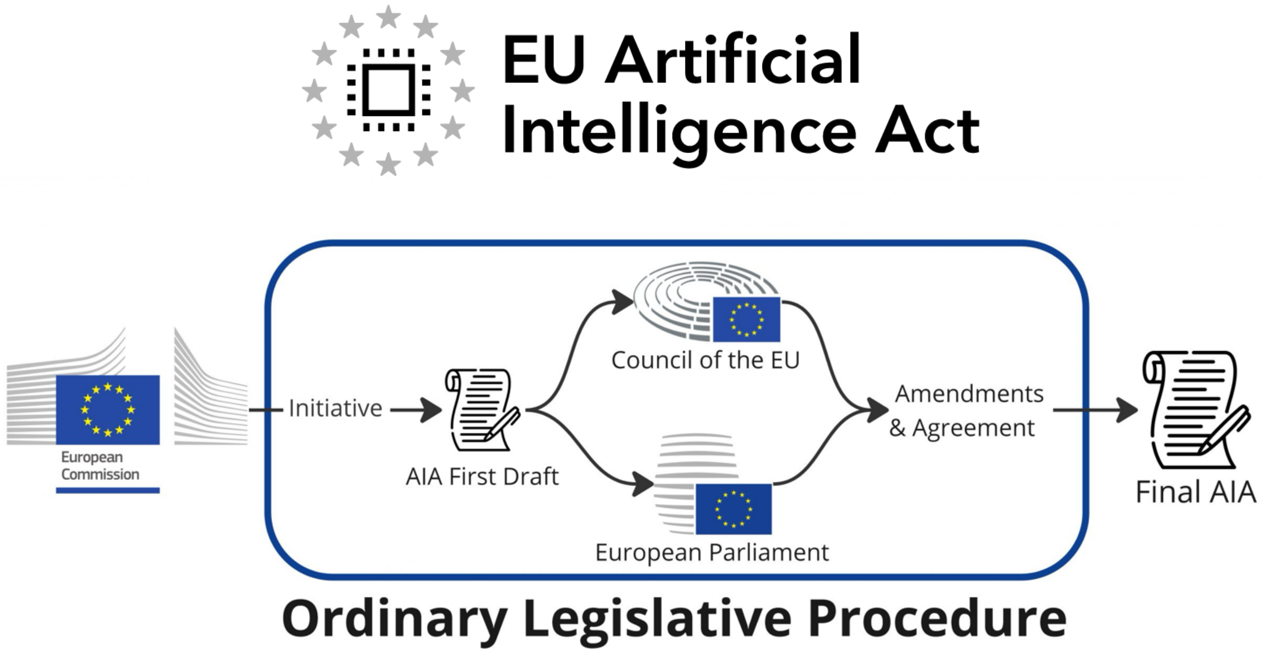
First Image of a Black Hole



reconstructed from massive amounts of radio signals using Interferometry
Multi-view Geometry in Fourier domain (like MRI scans)

https://science.nasa.gov/resource/first-image-of-a-black-hole/

Ethics and Governance



https://artificialintelligenceact.eu/

AI (artificial intelligence)

Iran war heralds era of AI-powered bombing quicker than 'speed of thought'

Speed and scale of US military's AI war planning raises fears human decision-making may be sidelined

Academics say AI is collapsing the time required for military decision-making. Photograph: Majid Asgaripour/Reuters

Robert Booth *and* Dan Milmo

Tue 3 Mar 2026 06.00 GMT

Visual General Intelligence

-Vision Research Toward the AGI Era-

CVPR 2026 Workshop

Invited Speakers

Robert Geirhos
Google DeepMind

View Profile

Aditi Raghunathan
Carnegie Mellon University

View Profile

Matt Deitke
Meta Superintelligence Lab

View Profile

Kristen Grauman
University of Texas, Austin / Meta

View Profile

Yuki M. Asano
University of Technology Nuremberg

View Profile

Alexei A. Efros
University of California, Berkeley

View Profile

Jamie Shotton
Wayve

View Profile

Kaiming He
Massachusetts Institute of Technology / Google DeepMind

View Profile

Andrea Vedaldi
University of Oxford / Meta

View Profile

CVPR JUNE 3-7, 2026 DENVER COLORADO

VGI WORKSHOP

Summary – Applications & Challenges

- The Era of Large Models
  - Multimodal Language Models (MLLMs)
  - In-context learning
  - Image and video generation, interactive world generation
  - Segment Anything (SAM), SAM 3D
- Expanding Horizons
  - Agentic AI
  - Physical AI
  - AI for Science
  - Ethics and governance

Codex, Why You Are So Slow?

An AI-generated song for vibe coders
by Johnny Vibe Building

- Grok for fast & smooth video motion
- Midjourney for nice Niji 7 anime style
- Kimi for aesthetic prompts
- Suno v5: Amazing!
- OpenAI: Please make GPT 5.3 xhigh lightning fast! 😂

https://www.youtube.com/watch?v=bSzfGfPLgsg